
Bilinear Bandits with Partially Observable Features

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Abstract

We study bilinear bandits with partially observable features on both the user and item sides. In each of T rounds, the learner selects a user-item pair and observes only the reward for the chosen pair. The reward model is bilinear in the user and item features with an unknown parameter matrix. Existing work commonly reduces this problem to a linear bandit via Kronecker-product features, increasing dimensionality and losing bilinear structure. We propose BiRoLF, an algorithm robust to latent features, which directly leverages the bilinear structure without such linearization. It augments observed feature spaces on both sides with orthogonal complement bases and employs doubly robust (DR) estimation to impute rewards for unselected pairs, constructing matrix-valued pseudo-rewards. We estimate the effective parameter using a bilinear DR-Lasso estimator, which promotes sparsity in components orthogonal to observed features. BiRoLF achieves a $\tilde{O}(\sqrt{(d_x + d_{h_x})(d_y + d_{h_y})T})$ regret bound, where d_x and d_y are observable feature dimensions, and d_{h_x} and d_{h_y} denote effective orthogonal-complement dimensions. By exploiting the induced block-diagonal Gram structure, BiRoLF performs an exact blockwise update that preserves the full DR-Lasso objective. Numerical experiments show strong regret performance and computational gains.

1. Introduction

Linear bandits provide a principled framework for sequential decision making with side information, where the expected reward is modeled as a linear function of an observed feature vector (Dani et al., 2008; Abbasi-Yadkori et al., 2011). Many practical applications, however, involve inter-

actions between two entities, each with its own feature representation. Examples include recommendation systems that match users and items, ride-sharing platforms that match riders and drivers, and matching platforms that pair participants based on mutual suitability. In such settings, rewards depend not only on the individual features of each entity but also on their interaction. The bilinear bandit model captures this structure by parameterizing rewards through a bilinear form of two feature vectors. Prior work has leveraged additional structure in bilinear bandits to obtain regret guarantees, including low-rank reward matrices (Jun et al., 2019) and favorable action-space geometry (Jang et al., 2021).

Despite these advances, existing bilinear bandit formulations (Jun et al., 2019; Jang et al., 2021; Lu et al., 2021; Kang et al., 2022) typically assume that the feature representations of both entities are fully observed. This assumption is often unrealistic in practice. Available descriptors may capture only coarse demographic, behavioral, or contextual information, while reward-relevant factors such as latent preferences, compatibility, or suitability remain unobserved. As a result, the learner must make decisions based on incomplete feature information. This form of persistent partial observability is not merely a reduction in information: when latent features influence rewards, algorithms that rely solely on observed features can be fundamentally misspecified and may incur linear regret.

Recently, Kim et al. (2025) studied partial observability in linear bandits and proposed a robust learning framework based on orthogonal augmentation and doubly robust (DR) estimation (Bang & Robins, 2005; Dudík et al., 2011). A natural approach to extending their framework to bilinear bandits is to vectorize the two-sided features through a Kronecker product and reduce the problem to a partially observable linear bandit. However, this reduction destroys the bilinear structure, resulting in substantially higher computational costs and preventing the use of algorithmic techniques developed specifically for bilinear bandits. These limitations motivate the need for a dedicated framework for bilinear bandits with partially observable features.

In this paper, we study bilinear bandits under two-sided partial observability. At each round, the learner selects a pair from a fixed two-sided action pool and observes only the reward of the chosen pair. The learner has access to observed

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descriptors for both sides, but some reward-relevant components of each descriptor remain latent throughout learning. This setting captures decision problems in which pairwise rewards are driven by both observed information and unobserved factors. The main challenge is that the learner must account for latent reward components without abandoning the bilinear structure that makes the model statistically and computationally tractable.

We propose a bilinear learning framework for bandits with two-sided partially observable features. The key idea is to preserve the bilinear structure by augmenting the observed feature spaces on both sides with orthogonal-complement directions. This yields a matrix-valued effective parameter, rather than the vector-valued parameter induced by a naive Kronecker-product reduction. The resulting formulation separates the observed and latent components across the two feature spaces and induces an effective sparsity structure over the orthogonal-complement directions. To estimate this parameter from bandit feedback, we develop a bilinear DR Lasso estimator.

A further advantage of the proposed formulation is computational. Although the augmented bilinear model could in principle be optimized as a large vectorized linear model, doing so would ignore the block structure induced by the two-sided augmentation. We show that the corresponding Gram matrix admits a block-diagonal structure, which allows the full bilinear DR-Lasso objective to be decomposed into blockwise subproblems without changing the set of global minimizers. Thus, our method is not merely a heuristic acceleration of the naive reduction, but an exact reformulation that preserves both the statistical structure and the computational advantages of the bilinear model.

We summarize our main contribution as follows:

- We introduce BiRoLF, a provably efficient algorithm for bilinear bandits with two-sided partially observable features. The algorithm preserves the bilinear structure through two-sided orthogonal augmentation, yielding a matrix-valued effective parameter and pairwise DR pseudo-reward matrices over user–item pairs. We establish consistency of the resulting bilinear DR-Lasso estimator and obtain a regret guarantee of $\tilde{O}(\sqrt{(d_x + d_{h_x})(d_y + d_{h_y})T})$, where d_{h_x} and d_{h_y} denote the effective orthogonal-complement dimensions.
- We show that the augmented bilinear formulation yields an exact computational advantage, not merely a heuristic speedup. By exploiting the block-diagonal Gram structure induced by the two-sided orthogonal augmentation, we reformulate the full DR-Lasso objective into blockwise subproblems with the same global minimizers, reducing the per-iteration cost from $\mathcal{O}((MN)^2)$ under naive Kronecker-product lineariza-

tion to $\mathcal{O}(Nd_x^2 + Md_y^2)$.

- We empirically assess both the statistical and computational performance of BiRoLF. The evaluation compares cumulative regret, optimization time, and total execution time on synthetic and semi-synthetic recommendation instances, and includes bilinear baselines and ablations isolating the effects of DR estimation, two-sided augmentation, and blockwise optimization.

2. Related Works

In off-policy evaluation for contextual bandits, a substantial line of work is built on the inverse propensity scoring (IPS) and DR estimators, which combine reward modeling with propensity weighting (Dudík et al., 2011; Wang et al., 2017; Farajtabar et al., 2018). Related approaches adopt DR estimators and balancing/reweighting methods in online contextual bandits (Kim & Paik, 2019; Dimakopoulou et al., 2019; Kim et al., 2021; 2023b; 2025).

Unlike the literature on model misspecification (Ghosh et al., 2017; Krishnamurthy et al., 2021) where the mean reward deviates from the assumed model, we focus on a well-specified reward model with partially observable features, where the major challenge is information loss. A line of work studies contextual bandits in which contexts are imperfectly observed, including noisy feature observations and context uncertainty (Kim et al., 2023a; Wang et al., 2022), as well as settings with partially observable features (Kim et al., 2025). Another direction considers latent confounding in contextual bandits (Sen et al., 2017; Tennenholtz et al., 2021) and models latent structures such as hidden states or user clusters (Zhou & Brunskill, 2016; Hong et al., 2020; Nelson et al., 2022; Pal et al., 2023). Several works on partially observed contexts assume an explicit observation model and focus on recovering the latent contexts (Zeng et al., 2025). In contrast, we do not assume such generative structure and instead target robust online learning when only partial feature information is revealed.

These works on partial observability primarily focus on linear reward models. Most closely related is RoLF (Kim et al., 2025), which studies one-sided partially observable linear bandits and constructs a vector-valued effective parameter and vector-valued pseudo-rewards. In contrast, our setting involves a two-sided bilinear reward, leading to a matrix-valued effective parameter, pairwise pseudo-reward matrices over user–item pairs, and a bilinear DR-Lasso estimator.

Bilinear bandits, on the other hand, study settings where an action consists of a pair of arms from two entities and the expected reward is a bilinear function of their feature vectors. Existing bilinear bandit algorithms typically assume fully observed features and leverage additional structure, such as the low-rank structure of the unknown matrix parameter or

action-space geometry (Jun et al., 2019; Jang et al., 2021). Closely related work studies learning unknown low-rank matrix parameters under linear or generalized linear reward models (Lu et al., 2021; Kang et al., 2022). Variants also include graphical bilinear bandits, where interactions are defined over graph edges in best-arm identification (Rizk et al., 2021). Our work differs from prior bilinear-bandit lines by addressing persistent two-sided partial observability without assuming full feature observability or low-rank structure.

3. Preliminaries

3.1. Notation

For any integer $n \geq 1$, let $[n] = \{1, \dots, n\}$. For any vector $\mathbf{v}$, denote ℓ_1 , ℓ_2 and ℓ_∞ norms by $\|\mathbf{v}\|_1$, $\|\mathbf{v}\|_2$, and $\|\mathbf{v}\|_\infty$, respectively. For any matrix $\mathbf{M}$ and $p, q \in [1, \infty]$, let $\mathbf{r}_i$ be the i -th row vector of $\mathbf{M}$. Define a vector $\mathbf{m}$ such that its i -th element is $m_i = \|\mathbf{r}_i\|_p$, then denote $\|\mathbf{M}\|_{p,q} = \|\mathbf{m}\|_q$. For example, $\|\mathbf{M}\|_{2,2}$ is the same as the Frobenius norm of $\mathbf{M}$. For any positive definite matrix $\mathbf{D}$, denote the $\mathbf{D}$ -weighted ℓ_2 norm by $\|\mathbf{v}\|_{\mathbf{D}} := \sqrt{\mathbf{v}^\top \mathbf{D} \mathbf{v}}$. We use the dot product as the inner product, that is, for any vectors $\mathbf{v}_1, \mathbf{v}_2$, $\langle \mathbf{v}_1, \mathbf{v}_2 \rangle = \mathbf{v}_1^\top \mathbf{v}_2$. We denote $\text{Row}(\mathbf{M})$ as the row space of $\mathbf{M}$. For a matrix $\mathbf{M}$, $\text{Row}(\mathbf{M})$ denotes its row space and $\text{Row}(\mathbf{M})^\perp$ its orthogonal complement.

3.2. Problem Formulation

We consider a bilinear bandit with partially observable features on both sides. The arm set is $\mathcal{A} = [M] \times [N]$. Each arm $a = (i, j) \in \mathcal{A}$ corresponds to a user-item pair.

Features. Each user $i \in [M]$ has a full feature vector $\mathbf{x}_i^* \in \mathbb{R}^{d_x^*}$ and each item $j \in [N]$ has a full feature vector $\mathbf{y}_j^* \in \mathbb{R}^{d_y^*}$, decomposed as

$$\mathbf{x}_i^* = [\mathbf{x}_i^\top \quad \mathbf{u}_i^\top]^\top, \quad \mathbf{y}_j^* = [\mathbf{y}_j^\top \quad \mathbf{v}_j^\top]^\top$$

where $\mathbf{x}_i \in \mathbb{R}^{d_x}$ and $\mathbf{y}_j \in \mathbb{R}^{d_y}$ are observable, while $\mathbf{u}_i$ and $\mathbf{v}_j$ are latent and never observed. The learner observes $\{\mathbf{x}_i\}_{i=1}^M$ and $\{\mathbf{y}_j\}_{j=1}^N$. Throughout, $\{\mathbf{x}_i^*, \mathbf{y}_j^*\}$ are fixed over time.

Reward model. There is an unknown parameter matrix $\Theta_* \in \mathbb{R}^{d_x^* \times d_y^*}$. At each round t , the learner selects an arm $a_t = (i_t, j_t) \in \mathcal{A}$ and observes only the reward $r_{a_t, t}$ of the selected pair. For any arm $a = (i, j) \in \mathcal{A}$, we model the (potential) reward by

$$r_{a, t} = \mathbf{x}_i^{*\top} \Theta_* \mathbf{y}_j^* + \epsilon_{a, t}.$$

where $\epsilon_{a, t}$ is the random noise. Let $\mathcal{F}_t := \sigma(\{a_s, r_{a_s, s}\}_{s \leq t})$ denote the history up to round t including the internal randomness of the algorithm.

Objective and regret. The learner's objective is to maximize the cumulative expected reward while observing the observable features $\{\mathbf{x}_i\}_{i=1}^M$ and $\{\mathbf{y}_j\}_{j=1}^N$ and bandit feedback. Let $(i^*, j^*) \in \arg\max_{(i, j) \in \mathcal{A}} \mathbf{x}_i^{*\top} \Theta_* \mathbf{y}_j^*$ be an optimal arm (ties broken arbitrarily). We measure the performance of our algorithm by the cumulative pseudo-regret defined as

$$\text{Regret}(T) = \sum_{t=1}^T (\mathbf{x}_{i^*}^{*\top} \Theta_* \mathbf{y}_{j^*}^* - \mathbf{x}_{i_t}^{*\top} \Theta_* \mathbf{y}_{j_t}^*).$$

Assumptions. For the regret analysis, we work under the following standard assumptions.

Assumption 3.1 (Sub-Gaussian noise). The noise $\epsilon_{(i, j), t}$ is mean-zero and σ -sub-Gaussian given $\mathcal{F}_{t-1}$. For all $t \geq 1$, $a \in \mathcal{A}$ and $\lambda \in \mathbb{R}$,

$$\begin{aligned} \mathbb{E}[\epsilon_{a, t} \mid \mathcal{F}_{t-1}] &= 0, \\ \mathbb{E}[\exp(\lambda \epsilon_{a, t}) \mid \mathcal{F}_{t-1}] &\leq \exp(\lambda^2 \sigma^2 / 2). \end{aligned}$$

Assumption 3.2 (Bounded mean rewards). The expected reward is uniformly bounded. For all $(i, j) \in [M] \times [N]$,

$$|\mathbf{x}_i^{*\top} \Theta_* \mathbf{y}_j^*| \leq 1.$$

Assumption 3.3 (Full row-rank of observed feature matrices). Let $\mathbf{X} := [\mathbf{x}_1, \dots, \mathbf{x}_M] \in \mathbb{R}^{d_x \times M}$ and $\mathbf{Y} := [\mathbf{y}_1, \dots, \mathbf{y}_N] \in \mathbb{R}^{d_y \times N}$. Assume $\text{rank}(\mathbf{X}) = d_x$ and $\text{rank}(\mathbf{Y}) = d_y$ (equivalently, $\mathbf{X}\mathbf{X}^\top \succ 0$ and $\mathbf{Y}\mathbf{Y}^\top \succ 0$).

Remark 3.4. Assumption 3.3 is standard in partially observable bandits that employ orthogonal-basis augmentation. The row-space projection used by Kim et al. (2025) requires $\mathbf{X}\mathbf{X}^\top$ to be invertible. If $\text{rank}(\mathbf{X}) < d_x$, then the observed representation contains redundant coordinates. One may reduce the effective observed dimension to $\text{rank}(\mathbf{X})$ by changing coordinates on the observed row space, for instance via SVD, and express the unknown matrix in the reduced coordinates without changing any bilinear mean rewards. The same reduction applies to $\mathbf{Y}$ when $\text{rank}(\mathbf{Y}) < d_y$. Thus, after this coordinate reduction, Assumption 3.3 holds with effective dimensions $\text{rank}(\mathbf{X})$ and $\text{rank}(\mathbf{Y})$.

4. Algorithm

In this section, we present our main algorithm BiRoLF (Algorithm 1) for addressing partial observability in a bilinear bandit. The algorithm proceeds by first constructing an augmented bilinear representation, then using DR pseudo-rewards to compensate for missing feedback, and finally updating a sparse estimator. We augment features using orthogonal-complement bases so that the mean reward can be represented by an augmented parameter Φ_* (Section 4.1). Feature augmentation induces a block structure in the augmented parameter, and the blocks involving the orthogonal-complement coordinates are typically sparse. Therefore, we

estimate the augmented parameter by ℓ_1 -regularization (Section 4.2). An obstacle is that the DR correction (Section 4.3) is computable only for matching events between the chosen action and a pseudo-action drawn from the coupling distribution. Section 4.4 introduces a resampling scheme that ensures these matching events occur sufficiently often with high probability. Finally, we update the main estimator $\hat{\Phi}_t$ using the accumulated pseudo-rewards (Section 4.5). We then develop the blockwise implementation in Section 5. Before turning to the augmented representation, we define the forced-exploration schedule, which ensures sufficient coverage for the estimator used below.

Forced exploration schedule. Following Kim et al. (2025), we maintain a set of exploration rounds $\mathfrak{E}_t \subseteq [t]$ that grows over time. Starting from $\mathfrak{E}_0 = \emptyset$, we update it recursively by

$$\mathfrak{E}_t := \begin{cases} \mathfrak{E}_{t-1} \cup \{t\}, & \text{if } |\mathfrak{E}_{t-1}| \leq C_e \log \left(\frac{2MNt^2}{\delta} \right), \\ \mathfrak{E}_{t-1}, & \text{otherwise,} \end{cases} \quad (1)$$

where C_e is a constant specified as

$$C_e = \frac{512\tilde{\sigma}_{\max,z}^2 MN(MN - 2 + p)}{3\tilde{\sigma}_{\min,z}^2(1 - p)}.$$

$\tilde{\sigma}_{\max,z}$ and $\tilde{\sigma}_{\min,z}$ are defined in Section 6. At rounds $t \in \mathfrak{E}_t$, the action is sampled uniformly; otherwise, we use the main action selection rule in Algorithm 1.

4.1. Feature Vector Augmentation

One challenge in the partially observed bandit problem is handling the unobserved reward component. In this section, we introduce a feature-augmentation method to represent this component. Let us define the feature matrices by stacking the user- and item-side features.

$$\begin{aligned} \mathbf{X} &= [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_M] \in \mathbb{R}^{d_x \times M} \\ \mathbf{U} &= [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_M] \in \mathbb{R}^{d_u \times M} \\ \mathbf{X}_* &= [\mathbf{X}^\top \quad \mathbf{U}^\top]^\top \in \mathbb{R}^{d_x^* \times M} \\ \mathbf{Y} &= [\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_N] \in \mathbb{R}^{d_y \times N} \\ \mathbf{V} &= [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_N] \in \mathbb{R}^{d_v \times N} \\ \mathbf{Y}_* &= [\mathbf{Y}^\top \quad \mathbf{V}^\top]^\top \in \mathbb{R}^{d_y^* \times N} \end{aligned}$$

where $d_x^* := d_x + d_u$ and $d_y^* := d_y + d_v$. Let $\mathbf{R}_t \in \mathbb{R}^{M \times N}$ be the reward matrix with entries $[\mathbf{R}_t]_{ij} = r_{(i,j),t}$. Under the bilinear model, the mean reward matrix satisfies

$$\mathbb{E}[\mathbf{R}_t \mid \mathcal{F}_{t-1}] = \mathbf{X}_*^\top \Theta_* \mathbf{Y}_*.$$

Writing Θ_* in observable and latent blocks, the mean reward matrix over user-item pairs consists of one observable-observable interaction and three interactions involving latent

Algorithm 1 Bilinear Robust to Latent Features (BiRoLF)

Require: Observed features $\{\mathbf{x}_i : i \in [M]\}$, $\{\mathbf{y}_j : j \in [N]\}$, coupling probability $p \in (1/2, 1)$, confidence parameter $\delta > 0$.

- 1: Initialize $\hat{\Phi}_0 \leftarrow \mathbf{0}_{M \times N}$ and the match counter $\Gamma \leftarrow 0$.
 - 2: Find an orthonormal basis $\{\mathbf{b}_{\mathbf{X}}^1, \mathbf{b}_{\mathbf{X}}^2, \dots, \mathbf{b}_{\mathbf{X}}^{M-d_x}\} \subset \text{Row}(\mathbf{X})^\perp$ and construct the rotated user-side features $\{\tilde{\mathbf{x}}_i : i \in [M]\}$. Analogously, construct $\{\tilde{\mathbf{y}}_j : j \in [N]\}$ for the item-side features.
 - 3: **for** $t = 1, \dots, T$ **do**
 - 4: Update the exploration set $\mathfrak{E}_t$ as (1).
 - 5: **if** $t \in \mathfrak{E}_t$ **then**
 - 6: Sample $\hat{a}_t = (\hat{i}_t, \hat{j}_t) \sim \text{Unif}([M] \times [N])$.
 - 7: **else**
 - 8: $\hat{a}_t \leftarrow \arg\max_{(i,j) \in [M] \times [N]} \tilde{\mathbf{x}}_i^\top \hat{\Phi}_{t-1} \tilde{\mathbf{y}}_j$.
 - 9: **end if**
 - 10: $\nu \leftarrow 0$.
 - 11: **while** $\nu < \rho_t$ **do**
 - 12: Sample $a_t = (i_t, j_t)$ from π_t and $\tilde{a}_t = (\tilde{i}_t, \tilde{j}_t)$ from $\phi_t(\cdot \mid a_t)$ as in (3) and (4).
 - 13: $\nu \leftarrow \nu + 1$.
 - 14: **if** $\tilde{a}_t = a_t$ **then**
 - 15: **break**
 - 16: **end if**
 - 17: **end while**
 - 18: Play $a_t = (i_t, j_t)$ and observe reward $r_{a_t,t}$.
 - 19: **if** $\tilde{a}_t = a_t$ **then**
 - 20: $\Gamma \leftarrow \Gamma + 1$.
 - 21: Update $\hat{\Phi}_t$ according to (2).
 - 22: Calculate pseudo-rewards $\tilde{r}_{(i,j),t}$ defined in (5).
 - 23: Update $\hat{\Phi}_t$ according to (7).
 - 24: **else**
 - 25: Set $\hat{\Phi}_t \leftarrow \hat{\Phi}_{t-1}$.
 - 26: **end if**
 - 27: **end for**
-

user or item features. The augmentation represents this matrix using the observed row spaces and their orthogonal complements, yielding the four blocks of the effective parameter Φ_* . We provide the detailed algebraic derivation in Appendix A.1. Let $\mathbf{B}_{\mathbf{X}^\perp}$ and $\mathbf{B}_{\mathbf{Y}^\perp}$ be orthonormal bases of $\text{Row}(\mathbf{X})^\perp$ and $\text{Row}(\mathbf{Y})^\perp$, respectively. Thus, there exists an effective parameter $\Phi_* \in \mathbb{R}^{M \times N}$ satisfying

$$\begin{aligned} \mathbb{E}[\mathbf{R}_t \mid \mathcal{F}_{t-1}] &= [\mathbf{X}^\top \quad \mathbf{B}_{\mathbf{X}^\perp}^\top] \Phi_* [\mathbf{Y}^\top \quad \mathbf{B}_{\mathbf{Y}^\perp}^\top]^\top \\ &= [\mathbf{X}^\top \quad \mathbf{B}_{\mathbf{X}^\perp}^\top] \begin{bmatrix} \Phi_{*}^{(o,o)} & \Phi_{*}^{(o,h)} \\ \Phi_{*}^{(h,o)} & \Phi_{*}^{(h,h)} \end{bmatrix} [\mathbf{Y}^\top \quad \mathbf{B}_{\mathbf{Y}^\perp}^\top]^\top. \end{aligned}$$

The block partition has row sizes d_x and $M - d_x$, and column sizes d_y and $N - d_y$. The block indices of Θ_* and Φ_* have different meanings, as illustrated in Figure 1. For Θ_* , the blocks correspond to observable and latent feature coordinates. In contrast, for Φ_* the “h” blocks correspond

	Y	V		Y	B_{Y[⊥]}
X	$\Theta_*^{(o,o)}$	$\Theta_*^{(o,u)}$	X	$\Phi_*^{(o,o)}$	$\Phi_*^{(o,h)}$
U	$\Theta_*^{(u,o)}$	$\Theta_*^{(u,u)}$	B_{X[⊥]}	$\Phi_*^{(h,o)}$	$\Phi_*^{(h,h)}$

(a) Block structure of Θ_* (b) Block structure of Φ_*

Figure 1. Comparison of block indices for Θ_* and Φ_* . For Φ_* , h denotes an orthogonal-complement coordinate rather than an original latent feature coordinate.

to orthogonal-complement coordinates spanned by $\mathbf{B}_{\mathbf{X}^\perp}$ and $\mathbf{B}_{\mathbf{Y}^\perp}$, namely directions orthogonal to $\text{Row}(\mathbf{X})$ and $\text{Row}(\mathbf{Y})$. Since our interest lies in evaluating the mean reward matrix $\mathbf{R}_t$ rather than estimating the parameter Θ_* itself, we treat Φ_* as the effective parameter and use the augmented feature matrices:

$$\tilde{\mathbf{X}} := [\mathbf{X}^\top \quad \mathbf{B}_{\mathbf{X}^\perp}^\top]^\top \quad \tilde{\mathbf{Y}} := [\mathbf{Y}^\top \quad \mathbf{B}_{\mathbf{Y}^\perp}^\top]^\top$$

We define the augmented feature vectors corresponding to matrices $\tilde{\mathbf{X}}$ and $\tilde{\mathbf{Y}}$ as

$$\tilde{\mathbf{x}}_i := [\mathbf{x}_i^\top \quad (\mathbf{e}_\mathbf{x})_i^\top \mathbf{B}_{\mathbf{X}^\perp}^\top]^\top \quad \tilde{\mathbf{y}}_j := [\mathbf{y}_j^\top \quad (\mathbf{e}_\mathbf{y})_j^\top \mathbf{B}_{\mathbf{Y}^\perp}^\top]^\top$$

where $(\mathbf{e}_\mathbf{x})_i$ and $(\mathbf{e}_\mathbf{y})_j$ are standard basis vectors of $\mathbb{R}^M$ and $\mathbb{R}^N$, respectively.

4.2. Sparsity and Lasso Regularization

The augmented parameter Φ_* need not be dense in the orthogonal-complement blocks. On the user side, suppose that the component of the latent reward contribution lying outside $\text{Row}(\mathbf{X})$ is supported on only a small number of directions in the basis $\mathbf{B}_{\mathbf{X}^\perp}$. Then only the corresponding rows of $\Phi_*^{(h,o)}$ and $\Phi_*^{(h,h)}$ can be nonzero. Similarly, if the item-side latent contribution outside $\text{Row}(\mathbf{Y})$ is supported on a small number of directions in $\mathbf{B}_{\mathbf{Y}^\perp}$, then only the corresponding columns of $\Phi_*^{(o,h)}$ and $\Phi_*^{(h,h)}$ can be nonzero. This motivates estimating Φ_* by ℓ_1 -regularization, which promotes sparsity in the augmented effective parameter.

We measure the size of these active orthogonal-complement components by two quantities. Let d_{h_x} denote the number of nonzero row vectors across $\Phi_*^{(h,o)}$ and $\Phi_*^{(h,h)}$, and let d_{h_y} denote the number of nonzero column vectors across $\Phi_*^{(o,h)}$ and $\Phi_*^{(h,h)}$. These quantities enter the regret bound as the effective dimensions of the user- and item-side orthogonal-complement components.

4.3. Reward Imputation via Doubly Robust Estimation

In our bandit setting, only the reward of the selected action $a_t = (i_t, j_t)$ is observed at round t , while the other $MN - 1$ pair rewards are missing. To construct estimator updates over all user-item pairs, we use a doubly robust (DR) imputation scheme. The idea is to combine an outcome-model prediction with an inverse-propensity correction on

the observed reward, following the standard DR principle used in missing-data, off-policy evaluation, and contextual bandit settings (Bang & Robins, 2005; Dudík et al., 2011; Wang et al., 2017). In our setting, the action propensities and the coupling probabilities introduced below are controlled by the learner and hence known.

We first define the outcome model used for imputation.

$$\check{\Phi}_t := \arg\min_{\Phi} \sum_{\tau=1}^t (r_{a_\tau, \tau} - \tilde{\mathbf{x}}_{i_\tau}^\top \Phi \tilde{\mathbf{y}}_{j_\tau})^2 + 2\sigma \kappa_{\max, x} \kappa_{\max, y} \sqrt{2t \log \frac{2MNt^2}{\delta}} \|\Phi\|_{1,1}, \quad (2)$$

where $\kappa_{\max, x}$ and $\kappa_{\max, y}$ denote $\max_{i \in [M]} \|\tilde{\mathbf{x}}_i\|_\infty$ and $\max_{j \in [N]} \|\tilde{\mathbf{y}}_j\|_\infty$ respectively.

The learner chooses an action $a_\tau = (i_\tau, j_\tau)$ from an ϵ_τ -greedy distribution π_τ . Given the estimator from the previous round, define the ϵ_τ -greedy action as

$$(\hat{i}_\tau, \hat{j}_\tau) \in \arg\max_{(i,j) \in [M] \times [N]} \tilde{\mathbf{x}}_i^\top \hat{\Phi}_{\tau-1} \tilde{\mathbf{y}}_j.$$

Let $\epsilon_\tau := 1/\sqrt{\tau}$. Then π_τ is defined by

$$\pi_\tau(a) = \begin{cases} 1 - \epsilon_\tau, & a = (\hat{i}_\tau, \hat{j}_\tau) \\ \epsilon_\tau / (MN - 1), & a \neq (\hat{i}_\tau, \hat{j}_\tau). \end{cases} \quad (3)$$

As τ increases, the exploration rate ϵ_τ decays and the distribution π_τ becomes concentrated on $(\hat{i}_\tau, \hat{j}_\tau)$. Consequently, raw inverse-propensity weights for non-greedy actions can become large, increasing the variance of the correction term. To avoid this instability, we introduce a pseudo-action $\tilde{a}_\tau = (\tilde{i}_\tau, \tilde{j}_\tau)$ sampled from a coupling distribution with a constant match probability.

Given the action $a_\tau = (i_\tau, j_\tau)$, we sample a pseudo-action $\tilde{a}_\tau = (\tilde{i}_\tau, \tilde{j}_\tau)$ from the following coupling distribution ϕ_τ .

$$\phi_\tau(a | a_\tau) = \begin{cases} p, & a = a_\tau \\ (1-p)/(MN-1), & a \neq a_\tau, \end{cases} \quad (4)$$

where $p \in (1/2, 1)$ is a hyperparameter.

We use the pseudo-action to form pseudo-rewards only on match rounds $\mathcal{M}_\tau = \{\tilde{a}_\tau = a_\tau\}$. Note the distinction between the indices τ and t . At time t , for each matched round $\tau \leq t$, we update the pseudo-reward for every pair (i, j) using the current imputation parameter $\check{\Phi}_t$ as follows:

$$\tilde{r}_{(i,j), \tau} \leftarrow \tilde{\mathbf{x}}_i^\top \check{\Phi}_t \tilde{\mathbf{y}}_j + \frac{\mathbb{1}\{(i,j) = a_\tau\}}{p} (r_{a_\tau, \tau} - \tilde{\mathbf{x}}_{i_\tau}^\top \check{\Phi}_t \tilde{\mathbf{y}}_{j_\tau}). \quad (5)$$

On match rounds, the coupling probability satisfies $\phi_\tau(a_\tau | a_\tau) = p$, so the correction uses the bounded coupling weight $1/p$ rather than the potentially large raw inverse propensity

under π_τ . The first term imputes rewards for all user–item pairs, while the second term corrects the imputation using the actually observed reward of the selected pair. Thus, the pseudo-reward matrix provides updates for unchosen pairs as well as the chosen pair under partial feedback.

4.4. Resampling Pseudo-Actions

Pseudo-rewards in (5) are formed only on match rounds because the correction term uses the actually observed reward, which is available only for the selected action $a_\tau = (i_\tau, j_\tau)$. To increase the frequency of match rounds, we use a resampling procedure before playing an action. Specifically, we repeatedly draw a candidate action from π_τ and a pseudo-action from the coupling distribution until they match, up to ρ_τ trials. We choose the resampling budget ρ_τ as follows:

$$\rho_\tau = \left\lceil \frac{\log(2(\tau+1)^2/\delta)}{-\log(1-p)} \right\rceil. \quad (6)$$

Let $\mathcal{M}_\tau$ denote the event that a match is obtained within ρ_τ resampling trials. With the choice of ρ_τ in (6), we obtain the high-probability guarantee

$$\mathbb{P}\left(\bigcap_{\tau=1}^t \mathcal{M}_\tau\right) \geq 1 - \delta.$$

See Appendix A.2 for the proof. Consequently, with probability at least $1 - \delta$, a match is obtained within ρ_τ trials for every $\tau \leq t$, so the pseudo-rewards are computable for all rounds up to time t . The resampling budget grows only logarithmically in τ and $1/\delta$. This is not the main computational bottleneck, since the dominant cost comes from solving the estimator update, which is addressed in Section 5.

4.5. Main Estimator

Combining the pseudo-rewards and the sparsity regularization above, we define the main estimator $\hat{\Phi}_t$ as follows:

$$\hat{\Phi}_t = \underset{\Phi \in \mathbb{R}^{M \times N}}{\operatorname{argmin}} \left(\sum_{\tau=1}^t \mathbb{1}(\mathcal{M}_\tau) \sum_{(i,j) \in [M] \times [N]} (\tilde{r}_{(i,j),\tau} - \tilde{\mathbf{x}}_i^\top \Phi \tilde{\mathbf{y}}_j)^2 + \lambda_t \|\Phi\|_{1,1} \right), \quad (7)$$

where the indicator $\mathbb{1}\{\mathcal{M}_\tau\}$ restricts the objective to match rounds, since pseudo-rewards are formed only when $\mathcal{M}_\tau$ occurs. The regularization parameter is set to

$$\lambda_t := \frac{4\sigma \kappa_{\max,x} \kappa_{\max,y}}{p} \sqrt{2t \log \frac{2MNt^2}{\delta}}.$$

For later use, let $F_t(\Phi)$ denote the objective in (7).

5. Exact Blockwise DR-Lasso Solver

This section is computational in nature. We do not introduce a new estimator, but show how to solve the DR-Lasso update in (7) while preserving the bilinear structure. A naive implementation would vectorize each augmented pair feature as $\operatorname{vec}(\tilde{\mathbf{x}}_i \tilde{\mathbf{y}}_j^\top)$ and solve a Lasso problem in dimension MN . In contrast, the two-sided orthogonal augmentation makes the user-side and item-side Gram matrices block diagonal. Consequently, the quadratic part of the DR-Lasso objective has no cross-block interactions among the four blocks (o, o) , (o, h) , (h, o) , (h, h) of Φ , while the entrywise ℓ_1 penalty is also block-separable. Therefore, the full DR-Lasso objective can be minimized exactly by solving the four blockwise subproblems independently and assembling their minimizers.

5.1. Blockwise Reformulation of the DR-Lasso Objective

Pseudo-reward aggregation. Fix an update time t at which at least one matched round has occurred. For each matched round $\tau \leq t$, let $a_\tau = (i_\tau, j_\tau)$. The pseudo-reward matrix is computed using the current imputation parameter $\check{\Phi}_t$. To keep notation light, we suppress this dependence on t and write

$$\tilde{\mathbf{R}}_\tau = [\tilde{r}_{(i,j),\tau}]_{i,j} = \tilde{\mathbf{X}}^\top \check{\Phi}_t \tilde{\mathbf{Y}} + \alpha_\tau (\mathbf{e}_x)_{i_\tau} (\mathbf{e}_y)_{j_\tau}^\top,$$

where $(\mathbf{e}_x)_i \in \mathbb{R}^M$ and $(\mathbf{e}_y)_j \in \mathbb{R}^N$ denote the i -th and j -th standard basis vectors, respectively, and

$$\alpha_\tau = \frac{1}{p} (r_{a_\tau,\tau} - \tilde{\mathbf{x}}_{i_\tau}^\top \check{\Phi}_t \tilde{\mathbf{y}}_{j_\tau}).$$

The first term imputes rewards for all user–item pairs, while the second term applies the inverse-coupling correction to the observed pair on the matched round.

Define the matrix statistic

$$\mathbf{S}_\tau := \sum_{(i,j) \in \mathcal{A}} \tilde{\mathbf{x}}_i \tilde{r}_{(i,j),\tau} \tilde{\mathbf{y}}_j^\top = \tilde{\mathbf{X}} \tilde{\mathbf{R}}_\tau \tilde{\mathbf{Y}}^\top.$$

The cumulative sufficient statistics are

$$\Gamma_t := \sum_{\tau=1}^t \mathbb{1}(\mathcal{M}_\tau), \quad \mathbf{B}_t := \frac{1}{\Gamma_t} \sum_{\tau=1}^t \mathbb{1}(\mathcal{M}_\tau) \mathbf{S}_\tau.$$

Block-diagonal Gram structure. Define the full Gram matrices of the augmented feature vectors by

$$\mathbf{G}_\mathbf{X} := \sum_{i=1}^M \tilde{\mathbf{x}}_i \tilde{\mathbf{x}}_i^\top, \quad \mathbf{G}_\mathbf{Y} := \sum_{j=1}^N \tilde{\mathbf{y}}_j \tilde{\mathbf{y}}_j^\top.$$

By construction, the augmentation bases $\mathbf{B}_{\mathbf{X}^\perp}$ and $\mathbf{B}_{\mathbf{Y}^\perp}$ have orthonormal columns and are orthogonal to the ob-

served row spaces. Consequently,

$$\mathbf{G}_\mathbf{X} = \begin{bmatrix} \mathbf{G}_{\mathbf{X},o} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{M-d_x} \end{bmatrix}, \quad \mathbf{G}_\mathbf{Y} = \begin{bmatrix} \mathbf{G}_{\mathbf{Y},o} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{N-d_y} \end{bmatrix}, \quad (8)$$

where $\mathbf{G}_{\mathbf{X},o} = \sum_{i=1}^M \mathbf{x}_i \mathbf{x}_i^\top$ and $\mathbf{G}_{\mathbf{Y},o} = \sum_{j=1}^N \mathbf{y}_j \mathbf{y}_j^\top$.

Expanding the squared Frobenius loss in (7) gives

$$\frac{F_t(\Phi)}{\Gamma_t} = \text{tr}(\Phi^\top \mathbf{G}_\mathbf{X} \Phi \mathbf{G}_\mathbf{Y}) - 2 \langle \mathbf{B}_t, \Phi \rangle_F + \bar{\lambda}_t \|\Phi\|_{1,1} + c_t,$$

where $\bar{\lambda}_t := \lambda_t / \Gamma_t$, and c_t is independent of Φ .

Four-block decomposition. Partition Φ conformably with (8) as

$$\Phi = \begin{bmatrix} \Phi^{(o,o)} & \Phi^{(o,h)} \\ \Phi^{(h,o)} & \Phi^{(h,h)} \end{bmatrix}.$$

Here o denotes observed row-space coordinates and h denotes orthogonal-complement coordinates. The h blocks are not recovered latent features, but basis coordinates used to represent latent reward components. Using (8), the quadratic term decomposes as

$$\begin{aligned} \text{tr}(\Phi^\top \mathbf{G}_\mathbf{X} \Phi \mathbf{G}_\mathbf{Y}) &= \text{tr} \left((\Phi^{(o,o)})^\top \mathbf{G}_{\mathbf{X},o} \Phi^{(o,o)} \mathbf{G}_{\mathbf{Y},o} \right) \\ &+ \sum_{k=1}^{N-d_y} (\Phi_{:,k}^{(o,h)})^\top \mathbf{G}_{\mathbf{X},o} \Phi_{:,k}^{(o,h)} \\ &+ \sum_{\ell=1}^{M-d_x} \Phi_{\ell,:}^{(h,o)} \mathbf{G}_{\mathbf{Y},o} (\Phi_{\ell,:}^{(h,o)})^\top \\ &+ \|\Phi^{(h,h)}\|_F^2. \end{aligned} \quad (9)$$

Blockwise objectives. For $\Theta \in \mathbb{R}^{d_x \times d_y}$, define

$$\mathcal{Q}(\Theta) := \text{tr}(\Theta^\top \mathbf{G}_{\mathbf{X},o} \Theta \mathbf{G}_{\mathbf{Y},o}) = \langle \mathbf{G}_{\mathbf{X},o} \Theta \mathbf{G}_{\mathbf{Y},o}, \Theta \rangle_F.$$

For the off-diagonal blocks, write

$$\beta_k := \Phi_{:,k}^{(o,h)} \in \mathbb{R}^{d_x}, \quad \gamma_\ell := (\Phi_{\ell,:}^{(h,o)})^\top \in \mathbb{R}^{d_y}.$$

Using (9) and the block separability of the linear term and the entrywise ℓ_1 penalty, define

$$\begin{aligned} f_t^{(o,o)}(\Theta) &= \mathcal{Q}(\Theta) - 2 \langle \mathbf{B}_t^{(o,o)}, \Theta \rangle_F + \bar{\lambda}_t \|\Theta\|_{1,1}, \\ f_{t,k}^{(o,h)}(\beta) &= \beta^\top \mathbf{G}_{\mathbf{X},o} \beta - 2 (\mathbf{B}_t^{(o,h)})_{:,k}^\top \beta + \bar{\lambda}_t \|\beta\|_1, \\ f_{t,\ell}^{(h,o)}(\gamma) &= \gamma^\top \mathbf{G}_{\mathbf{Y},o} \gamma - 2 (\mathbf{B}_t^{(h,o)})_{\ell,:}^\top \gamma + \bar{\lambda}_t \|\gamma\|_1, \\ f_t^{(h,h)}(\Psi) &= \|\Psi\|_F^2 - 2 \langle \mathbf{B}_t^{(h,h)}, \Psi \rangle_F + \bar{\lambda}_t \|\Psi\|_{1,1}. \end{aligned}$$

Combining the four blockwise objectives, define

$$\begin{aligned} H_t(\Phi) &:= f_t^{(o,o)}(\Phi^{(o,o)}) + \sum_{k=1}^{N-d_y} f_{t,k}^{(o,h)}(\Phi_{:,k}^{(o,h)}) \\ &+ \sum_{\ell=1}^{M-d_x} f_{t,\ell}^{(h,o)}((\Phi_{\ell,:}^{(h,o)})^\top) + f_t^{(h,h)}(\Phi^{(h,h)}). \end{aligned} \quad (10)$$

Theorem 5.1 (Exact blockwise decomposition and global optimality). *Fix a round t with $\Gamma_t \geq 1$. Let F_t be the DR-Lasso objective in (7), and let H_t be defined in (10). Then there exists a constant c_t independent of Φ such that*

$$\frac{1}{\Gamma_t} F_t(\Phi) = H_t(\Phi) + c_t.$$

Consequently,

$$\underset{\Phi}{\operatorname{argmin}} F_t(\Phi) = \underset{\Phi}{\operatorname{argmin}} H_t(\Phi).$$

Moreover, H_t is separable across the four blocks (o,o) , (o,h) , (h,o) , (h,h) of Φ . Hence, assembling minimizers of the four blockwise subproblems yields a global minimizer of the original DR-Lasso objective.

The proof is deferred to Appendix B. By Theorem 5.1, it suffices to solve the four blockwise subproblems independently and assemble their minimizers. The variables and solvers of these subproblems are summarized in Table 1.

Solving the blockwise subproblems. The (o,o) block is a matrix Lasso problem and can be solved by proximal-gradient methods such as FISTA (Beck & Teboulle, 2009). Let

$$g_t^{(o,o)}(\Theta) := \mathcal{Q}(\Theta) - 2 \langle \mathbf{B}_t^{(o,o)}, \Theta \rangle_F$$

be the smooth part of $f_t^{(o,o)}$. Then

$$\nabla g_t^{(o,o)}(\Theta) = 2 \mathbf{G}_{\mathbf{X},o} \Theta \mathbf{G}_{\mathbf{Y},o} - 2 \mathbf{B}_t^{(o,o)}.$$

Each of the (o,h) and (h,o) blocks decomposes into independent vector Lasso problems. Let $g_{t,k}^{(o,h)}$ and $g_{t,\ell}^{(h,o)}$ denote the smooth parts of $f_{t,k}^{(o,h)}$ and $f_{t,\ell}^{(h,o)}$, respectively. For each $k \in [N - d_y]$ and $\ell \in [M - d_x]$, their gradients are

$$\begin{aligned} \nabla g_{t,k}^{(o,h)}(\beta_k) &= 2 \mathbf{G}_{\mathbf{X},o} \beta_k - 2 (\mathbf{B}_t^{(o,h)})_{:,k}, \\ \nabla g_{t,\ell}^{(h,o)}(\gamma_\ell) &= 2 \mathbf{G}_{\mathbf{Y},o} \gamma_\ell - 2 (\mathbf{B}_t^{(h,o)})_{\ell,:}^\top. \end{aligned}$$

Unlike the other blocks, the (h,h) block is entrywise separable. Hence, its minimizer is obtained in closed form via elementwise soft-thresholding:

$$\hat{\Phi}_t^{(h,h)} = \text{Soft}_{\bar{\lambda}_t/2}(\mathbf{B}_t^{(h,h)})$$

where $\text{Soft}_\tau(z) := \text{sign}(z) \max\{|z| - \tau, 0\}$ denotes the soft-thresholding operator applied elementwise.

Table 1. Four subproblems in the exact blockwise implementation of the DR-Lasso update.

Block	Variable	Subproblem	Solver	Main cost
(o, o)	$\Phi^{(o,o)} \in \mathbb{R}^{d_x \times d_y}$	matrix Lasso	proximal gradient	$\mathcal{O}(d_x^2 d_y + d_x d_y^2)$
(o, h)	$\{\beta_k\}_{k=1}^{N-d_y} \subseteq \mathbb{R}^{d_x}$	$N - d_y$ vector Lassos	proximal gradient	$\mathcal{O}((N - d_y)d_x^2)$
(h, o)	$\{\gamma_\ell\}_{\ell=1}^{M-d_x} \subseteq \mathbb{R}^{d_y}$	$M - d_x$ vector Lassos	proximal gradient	$\mathcal{O}((M - d_x)d_y^2)$
(h, h)	$\Phi^{(h,h)} \in \mathbb{R}^{(M-d_x) \times (N-d_y)}$	separable Lasso	soft-thresholding	$\mathcal{O}((M - d_x)(N - d_y))$

5.2. Computational Cost of the Blockwise Update

By Theorem 5.1, the blockwise update solves the same objective as (7). Its computational saving comes from applying proximal-gradient updates to four structured subproblems instead of the dense Kronecker-linearized problem. Under the naive Kronecker representation, each augmented pair feature is $\text{vec}(\tilde{\mathbf{x}}_i \tilde{\mathbf{y}}_j^\top) \in \mathbb{R}^{MN}$. A dense proximal-gradient step for the resulting ℓ_1 -regularized quadratic objective requires multiplying by an $(MN) \times (MN)$ Gram operator, costing $\mathcal{O}((MN)^2)$ per iteration. Entrywise soft-thresholding step costs only $\mathcal{O}(MN)$, so the gradient evaluation dominates.

The exact blockwise formulation avoids this dense Gram operator. For the (o, o) block, the dominant computation is $\mathbf{G}_{\mathbf{X},o} \Phi^{(o,o)} \mathbf{G}_{\mathbf{Y},o}$, which costs $\mathcal{O}(d_x^2 d_y + d_x d_y^2)$. The (o, h) block requires $\mathcal{O}(d_x^2)$ work for each of the $N - d_y$ vector Lasso problems, giving $\mathcal{O}((N - d_y)d_x^2)$. Similarly, the (h, o) block costs $\mathcal{O}((M - d_x)d_y^2)$. The (h, h) block is entrywise separable and is solved by one soft-thresholding step with cost $\mathcal{O}((M - d_x)(N - d_y))$. Thus, the gradient-based part of the blockwise update costs

$$\mathcal{O}(d_x^2 d_y + d_x d_y^2 + (N - d_y)d_x^2 + (M - d_x)d_y^2)$$

per iteration, with an additional closed-form $\mathcal{O}((M - d_x)(N - d_y))$ soft-thresholding step for the (h, h) block. This avoids the $\mathcal{O}(M^2 N^2)$ gradient step required by dense Kronecker linearization.

6. Regret Analysis

In this section, we present the regret analysis for Algorithm 1. We first introduce the following constants:

$$\begin{aligned}
 \tilde{\mathbf{z}}_{(i,j)} &:= \text{vec}(\tilde{\mathbf{x}}_i \tilde{\mathbf{y}}_j^\top), \\
 \kappa_{\max,x}^4 &:= \max_{i \in [M]} \max_{i' \in [M]} [(\mathbf{e}_x)_i^\top \tilde{\mathbf{x}}_{i'}]^4, \\
 \kappa_{\max,y}^4 &:= \max_{j \in [N]} \max_{j' \in [N]} [(\mathbf{e}_y)_j^\top \tilde{\mathbf{y}}_{j'}]^4, \\
 \tilde{\sigma}_{\max,z}^2 &:= \max_{k \in [MN]} (\mathbf{e}_z)_k^\top \sum_{(i,j) \in \mathcal{A}} \tilde{\mathbf{z}}_{(i,j)} \tilde{\mathbf{z}}_{(i,j)}^\top (\mathbf{e}_z)_k, \\
 \tilde{\sigma}_{\min,z}^2 &:= \lambda_{\min} \left(\sum_{i \in [M]} \sum_{j \in [N]} \tilde{\mathbf{z}}_{(i,j)} \tilde{\mathbf{z}}_{(i,j)}^\top \right).
 \end{aligned} \tag{11}$$

We use Lemma C.2 to control the deviation of the running Gram matrix and obtain an upper bound. Using the consistency of our DR bilinear Lasso estimator (Lemma C.3), we derive a regret bound for Algorithm 1 as follows.

Theorem 6.1 (Regret bound for Algorithm 1). *Let M and N be the number of users and items, respectively, and let T be the time horizon. Let $\tilde{\sigma}_{\max,z}, \tilde{\sigma}_{\min,z}, \kappa_{\max,x}, \kappa_{\max,y}$ be the constants defined in (11). For any $\delta \in (0, 1)$ and $p \in (1/2, 1)$, with probability at least $1 - 6\delta$, the cumulative regret of BiRoLF is bounded by*

$$\begin{aligned}
 \text{Regret}(T) &\leq \frac{1024MN(MN - 2 + p)\tilde{\sigma}_{\max,z}^2}{3(1 - p)\tilde{\sigma}_{\min,z}^2} \log \left(\frac{2MNT^2}{\delta} \right) \\
 &\quad + \frac{32\sigma\kappa_{\max,x}\kappa_{\max,y}}{\tilde{\sigma}_{\min,z}p} \times \\
 &\quad \sqrt{2(d_x + d_{h_x})(d_y + d_{h_y})T \log \left(\frac{2MNT^2}{\delta} \right)} \\
 &\quad + \sqrt{2T \log \left(\frac{2}{\delta} \right)} + 4\sqrt{T} + 2\delta.
 \end{aligned}$$

This theorem gives a regret guarantee for the structure-preserving bilinear formulation, where the estimator is analyzed directly as a matrix-valued DR-Lasso update rather than through a dense Kronecker-linearized reduction. The first term is the forced-exploration cost needed to control the Gram matrix, while the second term is the main statistical term over the effective bilinear dimension $(d_x + d_{h_x})(d_y + d_{h_y})$. The remaining terms are lower-order contributions from concentration, resampling, and failure-probability accounting.

The dependence on partial observability enters through d_{h_x} and d_{h_y} rather than the raw latent dimensions. If $\text{Row}(\mathbf{U}) \subseteq \text{Row}(\mathbf{X})$, then $d_{h_x} = 0$, and if $\text{Row}(\mathbf{V}) \subseteq \text{Row}(\mathbf{Y})$, then $d_{h_y} = 0$. Thus, when both latent spaces are absorbed by the observed row spaces, the leading term reduces to $\tilde{\mathcal{O}}(\sqrt{d_x d_y T})$. In the fully observable special case, this matches the standard finite-action linear-bandit scaling under Kronecker linearization up to logarithmic factors, where the ambient linear dimension is $d_x d_y$ and the number of arms is MN .

Case	User-side condition	Item-side condition
1	$\text{Row}(\mathbf{U}) \not\subseteq \text{Row}(\mathbf{X})$	$\text{Row}(\mathbf{V}) \not\subseteq \text{Row}(\mathbf{Y})$
2	$\text{Row}(\mathbf{U}) \not\subseteq \text{Row}(\mathbf{X})$	$\text{Row}(\mathbf{V}) \subseteq \text{Row}(\mathbf{Y})$
3	$\text{Row}(\mathbf{U}) \subseteq \text{Row}(\mathbf{X})$	$\text{Row}(\mathbf{V}) \not\subseteq \text{Row}(\mathbf{Y})$
4	$\text{Row}(\mathbf{U}) \subseteq \text{Row}(\mathbf{X})$	$\text{Row}(\mathbf{V}) \subseteq \text{Row}(\mathbf{Y})$

Figure 2. Four experimental cases. Rows are cases, and columns indicate whether the latent user-side and item-side row spaces are contained in the corresponding observed row spaces.

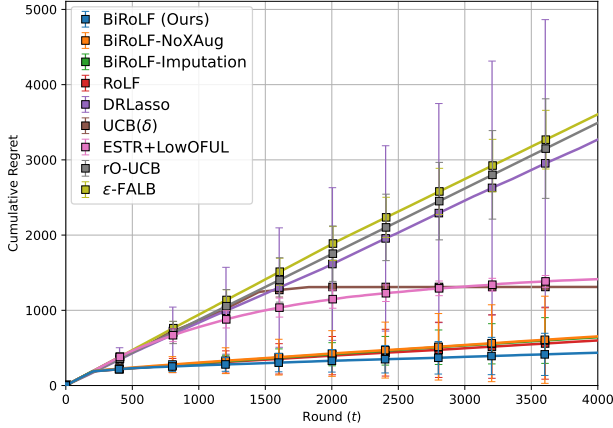


Figure 3. Case 1: Cumulative regret

7. Numerical Experiments

Experimental Setup. We evaluate BiRoLF on synthetic partially observable bilinear bandit instances, and report a semi-synthetic MovieLens experiment in Appendix E. Our goals are to assess regret under two-sided partial observability and quantify the computational gain from the blockwise update. We consider the four cases in Figure 2, which vary how much of the latent reward component can be represented through observed row spaces. We include RoLF (Kim et al., 2025) on Kronecker-product features as a strong partially observable baseline. We also include DRLasso (Kim & Paik, 2019) and UCB (Lattimore & Szepesvári, 2020) as sparse DR linear-bandit and feature-agnostic multi-armed bandit baselines. To compare with bilinear-bandit methods, we further include ESTR+LowOFUL (Jun et al., 2019), rO-UCB, and ϵ -FALB (Jang et al., 2021), run on the observed feature representation. We also report three ablations of BiRoLF: without blockwise updates, without user-side orthogonal augmentation, and without the DR correction. Cumulative regret is averaged over five independent trials. For computation, we report the main-estimator optimization time per trial and wall-clock time in Appendix E.

Experiment Results. For our experiments, we set $M = N = 20$, $d_x^* = d_y^* = 15$, and $d_x = d_y = 7$, so that the learner observes about half of the full feature dimen-

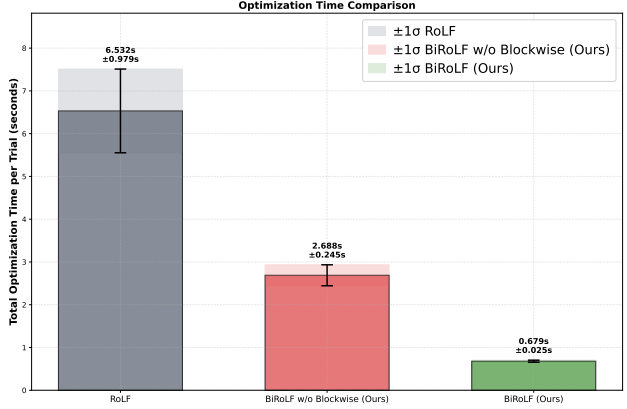


Figure 4. Case 1: Total optimization time per trial.

sion on each side. Figure 3 reports cumulative regret in Case 1, and Figure 4 compares main-estimator optimization time. BiRoLF maintains cumulative regret comparable to RoLF, while being faster than RoLF, which applies the partially observable linear-bandit method after Kronecker linearization. It is also faster than BiRoLF w/o Blockwise, which uses the same bilinear DR-Lasso estimator but solves the main update without the blockwise decomposition. This supports the computational benefit of the exact blockwise solver in Section 5. The cumulative-regret plots, wall-clock comparisons, and ablation results for the remaining cases are provided in Appendix E. Across these results, BiRoLF maintains regret comparable to the strongest partially observable baseline, while BiRoLF-NoXAug and BiRoLF-Imputation show the effect of removing user-side augmentation and the DR correction, respectively.

8. Conclusion

In this work, we studied bilinear bandits with two-sided partially observable features and proposed BiRoLF, a provably efficient algorithm that leverages the bilinear structure without Kronecker-product linearization. Our approach combines two-sided orthogonal augmentation with a doubly robust imputation scheme, and estimates the effective parameter via ℓ_1 -regularization to exploit sparsity in orthogonal-complement blocks. On the theoretical side, we established a regret guarantee of $\tilde{O}(\sqrt{(d_x + d_{h_x})(d_y + d_{h_y})T})$ under standard assumptions. On the computational side, we showed that the induced block-diagonal Gram structure yields an exact four-block decomposition of the DR-Lasso objective, enabling blockwise updates that reduce the cost compared to dense Kronecker linearization. Finally, our synthetic and semi-synthetic experiments show that BiRoLF maintains regret comparable to the strongest partially observable baseline while reducing optimization and wall-clock times. The ablations confirm the roles of augmentation, DR correction, and blockwise updates.

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Impact Statement

This paper presents work whose goal is to advance the theoretical foundations of machine learning, in particular, sequential decision-making under partial observability and bilinear reward models. While bandit algorithms may be applied in various real-world systems, such as recommendation or adaptive experimentation, the contributions of this work are primarily methodological and theoretical in nature. As such, we do not identify any specific broader societal or ethical impacts that warrant special discussion beyond those commonly associated with machine learning research.

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Appendix

A. Derivations for Feature Augmentation and Resampling

A.1. Two-Sided Feature Augmentation

We first collect the user- and item-side observed and latent features into matrices.

$$\begin{aligned} \mathbf{X} &= [\mathbf{x}_1, \dots, \mathbf{x}_M] \in \mathbb{R}^{d_x \times M}, & \mathbf{U} &= [\mathbf{u}_1, \dots, \mathbf{u}_M] \in \mathbb{R}^{d_u \times M}, \\ \mathbf{X}_* &= \begin{bmatrix} \mathbf{X} \\ \mathbf{U} \end{bmatrix} \in \mathbb{R}^{d_x^* \times M}, & d_x^* &:= d_x + d_u, \\ \mathbf{Y} &= [\mathbf{y}_1, \dots, \mathbf{y}_N] \in \mathbb{R}^{d_y \times N}, & \mathbf{V} &= [\mathbf{v}_1, \dots, \mathbf{v}_N] \in \mathbb{R}^{d_v \times N}, \\ \mathbf{Y}_* &= \begin{bmatrix} \mathbf{Y} \\ \mathbf{V} \end{bmatrix} \in \mathbb{R}^{d_y^* \times N}, & d_y^* &:= d_y + d_v. \end{aligned}$$

Let $\mathbf{R}_t \in \mathbb{R}^{M \times N}$ denote the potential reward matrix at round t , with (i, j) -entry $r_{(i,j),t}$. It can be written as

$$\begin{aligned} \mathbf{R}_t &= \mathbf{X}_*^\top \boldsymbol{\Theta}_* \mathbf{Y}_* + \mathcal{E}_t \\ &= \mathbf{X}^\top \boldsymbol{\Theta}_*^{(o,o)} \mathbf{Y} + \mathbf{U}^\top \boldsymbol{\Theta}_*^{(u,o)} \mathbf{Y} + \mathbf{X}^\top \boldsymbol{\Theta}_*^{(o,u)} \mathbf{V} + \mathbf{U}^\top \boldsymbol{\Theta}_*^{(u,u)} \mathbf{V} + \mathcal{E}_t, \end{aligned}$$

where $\mathcal{E}_t \in \mathbb{R}^{M \times N}$ is the noise matrix at time t .

We decompose the user- and item-side latent row-space components into their projections onto the observed row spaces and the corresponding orthogonal complements. Since $\mathbb{R}^M = \text{Row}(\mathbf{X}) \oplus \text{Row}(\mathbf{X})^\perp$ and $\mathbb{R}^N = \text{Row}(\mathbf{Y}) \oplus \text{Row}(\mathbf{Y})^\perp$, define the orthogonal projection matrices

$$\mathbf{P}_\mathbf{X} = \mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top)^{-1} \mathbf{X}, \quad \mathbf{P}_\mathbf{Y} = \mathbf{Y}^\top (\mathbf{Y}\mathbf{Y}^\top)^{-1} \mathbf{Y}.$$

Inserting $\mathbf{I}_M = \mathbf{P}_\mathbf{X} + (\mathbf{I}_M - \mathbf{P}_\mathbf{X})$ and $\mathbf{I}_N = \mathbf{P}_\mathbf{Y} + (\mathbf{I}_N - \mathbf{P}_\mathbf{Y})$ into the latent terms gives

$$\begin{aligned} \mathbf{R}_t &= \mathbf{X}^\top \left(\boldsymbol{\Theta}_*^{(o,o)} + (\mathbf{X}\mathbf{X}^\top)^{-1} \mathbf{X}\mathbf{U}^\top \boldsymbol{\Theta}_*^{(u,o)} + \boldsymbol{\Theta}_*^{(o,u)} \mathbf{V}\mathbf{Y}^\top (\mathbf{Y}\mathbf{Y}^\top)^{-1} + (\mathbf{X}\mathbf{X}^\top)^{-1} \mathbf{X}\mathbf{U}^\top \boldsymbol{\Theta}_*^{(u,u)} \mathbf{V}\mathbf{Y}^\top (\mathbf{Y}\mathbf{Y}^\top)^{-1} \right) \mathbf{Y} \\ &\quad + \mathbf{X}^\top \left(\boldsymbol{\Theta}_*^{(o,u)} + (\mathbf{X}\mathbf{X}^\top)^{-1} \mathbf{X}\mathbf{U}^\top \boldsymbol{\Theta}_*^{(u,u)} \right) \mathbf{V} (\mathbf{I}_N - \mathbf{P}_\mathbf{Y}) \\ &\quad + (\mathbf{I}_M - \mathbf{P}_\mathbf{X}) \mathbf{U}^\top \left(\boldsymbol{\Theta}_*^{(u,o)} + \boldsymbol{\Theta}_*^{(u,u)} \mathbf{V}\mathbf{Y}^\top (\mathbf{Y}\mathbf{Y}^\top)^{-1} \right) \mathbf{Y} \\ &\quad + (\mathbf{I}_M - \mathbf{P}_\mathbf{X}) \mathbf{U}^\top \boldsymbol{\Theta}_*^{(u,u)} \mathbf{V} (\mathbf{I}_N - \mathbf{P}_\mathbf{Y}) \\ &\quad + \mathcal{E}_t. \end{aligned}$$

Let

$$\mathbf{B}_{\mathbf{X}^\perp} = [\mathbf{b}_{\mathbf{X}^\perp,1}, \dots, \mathbf{b}_{\mathbf{X}^\perp,M-d_x}] \in \mathbb{R}^{M \times (M-d_x)}$$

and

$$\mathbf{B}_{\mathbf{Y}^\perp} = [\mathbf{b}_{\mathbf{Y}^\perp,1}, \dots, \mathbf{b}_{\mathbf{Y}^\perp,N-d_y}] \in \mathbb{R}^{N \times (N-d_y)}$$

be matrices whose columns form orthonormal bases of $\text{Row}(\mathbf{X})^\perp$ and $\text{Row}(\mathbf{Y})^\perp$, respectively. Since $\mathbf{P}_\mathbf{X}$ and $\mathbf{P}_\mathbf{Y}$ are the orthogonal projectors onto $\text{Row}(\mathbf{X})$ and $\text{Row}(\mathbf{Y})$, respectively, the complementary projectors are

$$\mathbf{I}_M - \mathbf{P}_\mathbf{X} = \mathbf{B}_{\mathbf{X}^\perp} \mathbf{B}_{\mathbf{X}^\perp}^\top, \quad \mathbf{I}_N - \mathbf{P}_\mathbf{Y} = \mathbf{B}_{\mathbf{Y}^\perp} \mathbf{B}_{\mathbf{Y}^\perp}^\top.$$

Define

$$\begin{aligned} \Phi_*^{(o,o)} &:= \boldsymbol{\Theta}_*^{(o,o)} + (\mathbf{X}\mathbf{X}^\top)^{-1} \mathbf{X}\mathbf{U}^\top \boldsymbol{\Theta}_*^{(u,o)} + \boldsymbol{\Theta}_*^{(o,u)} \mathbf{V}\mathbf{Y}^\top (\mathbf{Y}\mathbf{Y}^\top)^{-1} \\ &\quad + (\mathbf{X}\mathbf{X}^\top)^{-1} \mathbf{X}\mathbf{U}^\top \boldsymbol{\Theta}_*^{(u,u)} \mathbf{V}\mathbf{Y}^\top (\mathbf{Y}\mathbf{Y}^\top)^{-1}, \\ \Phi_*^{(o,h)} &:= \left(\boldsymbol{\Theta}_*^{(o,u)} + (\mathbf{X}\mathbf{X}^\top)^{-1} \mathbf{X}\mathbf{U}^\top \boldsymbol{\Theta}_*^{(u,u)} \right) \mathbf{V} \mathbf{B}_{\mathbf{Y}^\perp}, \\ \Phi_*^{(h,o)} &:= \mathbf{B}_{\mathbf{X}^\perp}^\top \mathbf{U}^\top \left(\boldsymbol{\Theta}_*^{(u,o)} + \boldsymbol{\Theta}_*^{(u,u)} \mathbf{V}\mathbf{Y}^\top (\mathbf{Y}\mathbf{Y}^\top)^{-1} \right), \\ \Phi_*^{(h,h)} &:= \mathbf{B}_{\mathbf{X}^\perp}^\top \mathbf{U}^\top \boldsymbol{\Theta}_*^{(u,u)} \mathbf{V} \mathbf{B}_{\mathbf{Y}^\perp}. \end{aligned} \tag{12}$$

Using the identities $\mathbf{I}_M - \mathbf{P}_\mathbf{X} = \mathbf{B}_{\mathbf{X}^\perp} \mathbf{B}_{\mathbf{X}^\perp}^\top$ and $\mathbf{I}_N - \mathbf{P}_\mathbf{Y} = \mathbf{B}_{\mathbf{Y}^\perp} \mathbf{B}_{\mathbf{Y}^\perp}^\top$, the reward matrix admits the augmented bilinear representation

$$\begin{aligned} \mathbf{R}_t &= [\mathbf{X}^\top \quad \mathbf{B}_{\mathbf{X}^\perp}^\top] \begin{bmatrix} \Phi_*^{(o,o)} & \Phi_*^{(o,h)} \\ \Phi_*^{(h,o)} & \Phi_*^{(h,h)} \end{bmatrix} \begin{bmatrix} \mathbf{Y} \\ \mathbf{B}_{\mathbf{Y}^\perp}^\top \end{bmatrix} + \mathcal{E}_t \\ &= [\mathbf{X}^\top \quad \mathbf{B}_{\mathbf{X}^\perp}^\top] \Phi_* \begin{bmatrix} \mathbf{Y} \\ \mathbf{B}_{\mathbf{Y}^\perp}^\top \end{bmatrix} + \mathcal{E}_t. \end{aligned}$$

The block definitions in (12) show that the unobservable reward components are represented through $\text{Row}(\mathbf{X})$, $\text{Row}(\mathbf{X})^\perp$, $\text{Row}(\mathbf{Y})$, and $\text{Row}(\mathbf{Y})^\perp$. For example, $\mathbf{U}^\top \Theta_*^{(u,o)} \mathbf{Y}$ is decomposed as

$$\mathbf{U}^\top \Theta_*^{(u,o)} \mathbf{Y} = \underbrace{\mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top)^{-1} \mathbf{X} \mathbf{U}^\top \Theta_*^{(u,o)} \mathbf{Y}}_{(i)} + \underbrace{(\mathbf{I}_M - \mathbf{P}_\mathbf{X}) \mathbf{U}^\top \Theta_*^{(u,o)} \mathbf{Y}}_{(ii)}$$

Term (i) is represented through the observed row spaces $\text{Row}(\mathbf{X})$ and $\text{Row}(\mathbf{Y})$, and therefore contributes to the (o, o) block. Term (ii) is represented through $\text{Row}(\mathbf{X})^\perp$ and $\text{Row}(\mathbf{Y})$, and therefore contributes to the (h, o) block. The same projection argument applies to the terms involving $\Theta_*^{(o,u)}$ and $\Theta_*^{(u,u)}$. Thus, the blocks of Θ_* are indexed by observed and latent feature coordinates, whereas the blocks of Φ_* are indexed by observed row-space and orthogonal-complement coordinates.

A.2. High-Probability Occurrence of Matching Events

Define $\mathcal{M}_\tau$ as the event where the pseudo action matches the chosen action before ρ_τ . Using the resampling times defined in (6), we can derive an inequality for $\mathbb{P}(\mathcal{M}_\tau)$ as follows:

$$\mathbb{P}(\mathcal{M}_\tau) = 1 - (1 - p)^{\rho_\tau} \geq 1 - \frac{\delta}{(\tau + 1)^2}.$$

Since $\sum_{\tau=1}^t \frac{1}{(\tau+1)^2} \leq \int_1^\infty \frac{1}{x^2} dx = 1$, the probability of the $\bigcap_{\tau=1}^t \mathcal{M}_\tau$ can be bounded as follows:

$$\mathbb{P}\left(\bigcap_{\tau=1}^t \mathcal{M}_\tau\right) = \mathbb{P}\left(\left(\bigcup_{\tau=1}^t \mathcal{M}_\tau^c\right)^c\right) = 1 - \mathbb{P}\left(\bigcup_{\tau=1}^t \mathcal{M}_\tau^c\right) \geq 1 - \sum_{\tau=1}^t \frac{\delta}{(\tau + 1)^2} \geq 1 - \delta$$

B. Proof of Theorem 5.1

Theorem 5.1 (restated). Fix a round t with $\Gamma_t := \sum_{\tau=1}^t \mathbb{1}(\mathcal{M}_\tau) \geq 1$. Let F_t be the objective function in (7), and let H_t be defined in (10). Then there exists a constant c_t independent of Φ such that

$$\frac{1}{\Gamma_t} F_t(\Phi) = H_t(\Phi) + c_t.$$

Consequently,

$$\underset{\Phi}{\operatorname{argmin}} F_t(\Phi) = \underset{\Phi}{\operatorname{argmin}} H_t(\Phi).$$

Moreover, H_t is separable across the four blocks (o, o) , (o, h) , (h, o) , (h, h) of Φ . Hence, assembling minimizers of the four blockwise subproblems yields a global minimizer of the original DR-Lasso objective.

Proof. Let the objective function in (7) be

$$F_t(\Phi) = \sum_{\tau=1}^t \mathbb{1}(\mathcal{M}_\tau) \sum_{(i,j) \in [M] \times [N]} (\tilde{r}_{(i,j),\tau} - \tilde{\mathbf{x}}_i^\top \Phi \tilde{\mathbf{y}}_j)^2 + \lambda_t \|\Phi\|_{1,1}.$$

Define $\tilde{\mathbf{R}}_\tau \in \mathbb{R}^{M \times N}$ as the pseudo-reward matrix whose (i, j) -entry equals $\tilde{r}_{(i,j),\tau}$. Let the augmented feature matrices $\tilde{\mathbf{X}} = [\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_M] \in \mathbb{R}^{M \times M}$ and $\tilde{\mathbf{Y}} = [\tilde{\mathbf{y}}_1, \dots, \tilde{\mathbf{y}}_N] \in \mathbb{R}^{N \times N}$. Then

$$\begin{aligned} \sum_{(i,j) \in [M] \times [N]} (\tilde{r}_{(i,j),\tau} - \tilde{\mathbf{x}}_i^\top \Phi \tilde{\mathbf{y}}_j)^2 &= \sum_{(i,j) \in [M] \times [N]} \left((\tilde{\mathbf{R}}_\tau)_{ij} - (\tilde{\mathbf{X}}^\top \Phi \tilde{\mathbf{Y}})_{ij} \right)^2 \\ &= \left\| \tilde{\mathbf{R}}_\tau - \tilde{\mathbf{X}}^\top \Phi \tilde{\mathbf{Y}} \right\|_F^2. \end{aligned}$$

Expanding the squared Frobenius norm yields

$$\begin{aligned}
 \|\tilde{\mathbf{R}}_\tau - \tilde{\mathbf{X}}^\top \Phi \tilde{\mathbf{Y}}\|_F^2 &= \|\tilde{\mathbf{R}}_\tau\|_F^2 - 2 \langle \tilde{\mathbf{R}}_\tau, \tilde{\mathbf{X}}^\top \Phi \tilde{\mathbf{Y}} \rangle_F + \|\tilde{\mathbf{X}}^\top \Phi \tilde{\mathbf{Y}}\|_F^2 \\
 &= \|\tilde{\mathbf{R}}_\tau\|_F^2 - 2 \operatorname{tr}(\tilde{\mathbf{R}}_\tau^\top \tilde{\mathbf{X}}^\top \Phi \tilde{\mathbf{Y}}) + \operatorname{tr}(\tilde{\mathbf{Y}}^\top \Phi^\top \tilde{\mathbf{X}} \tilde{\mathbf{X}}^\top \Phi \tilde{\mathbf{Y}}) \\
 &= \|\tilde{\mathbf{R}}_\tau\|_F^2 - 2 \operatorname{tr}(\tilde{\mathbf{Y}} \tilde{\mathbf{R}}_\tau^\top \tilde{\mathbf{X}}^\top \Phi) + \operatorname{tr}(\Phi^\top \tilde{\mathbf{X}} \tilde{\mathbf{X}}^\top \Phi \tilde{\mathbf{Y}} \tilde{\mathbf{Y}}^\top) \\
 &= \|\tilde{\mathbf{R}}_\tau\|_F^2 - 2 \langle \tilde{\mathbf{X}} \tilde{\mathbf{R}}_\tau \tilde{\mathbf{Y}}^\top, \Phi \rangle_F + \operatorname{tr}(\Phi^\top \mathbf{G}_X \Phi \mathbf{G}_Y),
 \end{aligned}$$

where $\mathbf{G}_X := \sum_{i=1}^M \tilde{\mathbf{x}}_i \tilde{\mathbf{x}}_i^\top = \tilde{\mathbf{X}} \tilde{\mathbf{X}}^\top$ and $\mathbf{G}_Y := \sum_{j=1}^N \tilde{\mathbf{y}}_j \tilde{\mathbf{y}}_j^\top = \tilde{\mathbf{Y}} \tilde{\mathbf{Y}}^\top$.

Ignoring constants independent of Φ , and using $\mathbf{S}_\tau := \tilde{\mathbf{X}} \tilde{\mathbf{R}}_\tau \tilde{\mathbf{Y}}^\top$, $\Gamma_t := \sum_{\tau=1}^t \mathbb{1}(\mathcal{M}_\tau)$, and $\mathbf{B}_t := \Gamma_t^{-1} \sum_{\tau=1}^t \mathbb{1}(\mathcal{M}_\tau) \mathbf{S}_\tau$, we obtain

$$\frac{1}{\Gamma_t} F_t(\Phi) = \bar{F}_t(\Phi) + c_t,$$

where c_t is independent of Φ and

$$\bar{F}_t(\Phi) := \operatorname{tr}(\Phi^\top \mathbf{G}_X \Phi \mathbf{G}_Y) - 2 \langle \mathbf{B}_t, \Phi \rangle_F + \frac{\lambda_t}{\Gamma_t} \|\Phi\|_{1,1}.$$

Next, we justify the block-diagonal forms in (8). Recall that the augmented feature matrices are

$$\tilde{\mathbf{X}} := [\mathbf{X}^\top \ \mathbf{B}_{X^\perp}^\top]^\top = \begin{bmatrix} \mathbf{X} \\ \mathbf{B}_{X^\perp}^\top \end{bmatrix}, \quad \tilde{\mathbf{Y}} := [\mathbf{Y}^\top \ \mathbf{B}_{Y^\perp}^\top]^\top = \begin{bmatrix} \mathbf{Y} \\ \mathbf{B}_{Y^\perp}^\top \end{bmatrix}.$$

By construction, the columns of $\mathbf{B}_{X^\perp}$ and $\mathbf{B}_{Y^\perp}$ form orthonormal bases of $\operatorname{Row}(\mathbf{X})^\perp$ and $\operatorname{Row}(\mathbf{Y})^\perp$, respectively. Hence,

$$\mathbf{X} \mathbf{B}_{X^\perp} = \mathbf{0}, \quad \mathbf{B}_{X^\perp}^\top \mathbf{B}_{X^\perp} = \mathbf{I}_{M-d_x}, \quad \mathbf{Y} \mathbf{B}_{Y^\perp} = \mathbf{0}, \quad \mathbf{B}_{Y^\perp}^\top \mathbf{B}_{Y^\perp} = \mathbf{I}_{N-d_y}.$$

Define $\mathbf{G}_{X,o} = \mathbf{X} \mathbf{X}^\top = \sum_{i=1}^M \mathbf{x}_i \mathbf{x}_i^\top$ and $\mathbf{G}_{Y,o} = \mathbf{Y} \mathbf{Y}^\top = \sum_{j=1}^N \mathbf{y}_j \mathbf{y}_j^\top$. Then, we obtain

$$\mathbf{G}_X = \tilde{\mathbf{X}} \tilde{\mathbf{X}}^\top = \begin{bmatrix} \mathbf{X} \mathbf{X}^\top & \mathbf{X} \mathbf{B}_{X^\perp}^\top \\ \mathbf{B}_{X^\perp}^\top \mathbf{X}^\top & \mathbf{B}_{X^\perp}^\top \mathbf{B}_{X^\perp} \end{bmatrix} = \begin{bmatrix} \mathbf{G}_{X,o} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{M-d_x} \end{bmatrix}.$$

Analogously, we have

$$\mathbf{G}_Y = \begin{bmatrix} \mathbf{G}_{Y,o} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{N-d_y} \end{bmatrix}.$$

With this structure, the quadratic term decomposes into four blocks of Φ :

$$\begin{aligned}
 \operatorname{tr}(\Phi^\top \mathbf{G}_X \Phi \mathbf{G}_Y) &= \operatorname{tr}\left((\Phi^{(o,o)})^\top \mathbf{G}_{X,o} \Phi^{(o,o)} \mathbf{G}_{Y,o}\right) + \sum_{k=1}^{N-d_y} (\Phi_{:,k}^{(o,h)})^\top \mathbf{G}_{X,o} \Phi_{:,k}^{(o,h)} \\
 &\quad + \sum_{\ell=1}^{M-d_x} (\Phi_{\ell,:}^{(h,o)}) \mathbf{G}_{Y,o} (\Phi_{\ell,:}^{(h,o)})^\top + \|\Phi^{(h,h)}\|_F^2
 \end{aligned}$$

Likewise, $\langle \mathbf{B}_t, \Phi \rangle_F$ is also block-separable:

$$\begin{aligned}
 \langle \mathbf{B}_t, \Phi \rangle_F &= \operatorname{tr}(\mathbf{B}_t^\top \Phi) \\
 &= \langle \mathbf{B}_t^{(o,o)}, \Phi^{(o,o)} \rangle_F + \langle \mathbf{B}_t^{(o,h)}, \Phi^{(o,h)} \rangle_F + \langle \mathbf{B}_t^{(h,o)}, \Phi^{(h,o)} \rangle_F + \langle \mathbf{B}_t^{(h,h)}, \Phi^{(h,h)} \rangle_F
 \end{aligned}$$

Moreover, the entrywise ℓ_1 penalty $\|\Phi\|_{1,1}$ is also block-separable:

$$\|\Phi\|_{1,1} = \|\Phi^{(o,o)}\|_{1,1} + \|\Phi^{(o,h)}\|_{1,1} + \|\Phi^{(h,o)}\|_{1,1} + \|\Phi^{(h,h)}\|_{1,1}.$$

Combining the above decompositions, the scaled objective $\bar{F}_t$ can be written as a sum of independent blockwise objectives:

$$\bar{F}_t(\Phi) = \text{tr} \left((\Phi^{(o,o)})^\top \mathbf{G}_{\mathbf{X},o} \Phi^{(o,o)} \mathbf{G}_{\mathbf{Y},o} \right) - 2 \left\langle \mathbf{B}_t^{(o,o)}, \Phi^{(o,o)} \right\rangle_{\text{F}} + \frac{\lambda_t}{\Gamma_t} \|\Phi^{(o,o)}\|_{1,1} \quad (\text{o,o})$$

$$+ \sum_{k=1}^{N-d_y} \left((\Phi_{:,k}^{(o,h)})^\top \mathbf{G}_{\mathbf{X},o} \Phi_{:,k}^{(o,h)} - 2 \left(\mathbf{B}_t^{(o,h)} \right)_{:,k}^\top \Phi_{:,k}^{(o,h)} + \frac{\lambda_t}{\Gamma_t} \|\Phi_{:,k}^{(o,h)}\|_1 \right) \quad (\text{o,h})$$

$$+ \sum_{\ell=1}^{M-d_x} \left((\Phi_{\ell,:}^{(h,o)}) \mathbf{G}_{\mathbf{Y},o} (\Phi_{\ell,:}^{(h,o)})^\top - 2 \left(\mathbf{B}_t^{(h,o)} \right)_{\ell,:} (\Phi_{\ell,:}^{(h,o)})^\top + \frac{\lambda_t}{\Gamma_t} \|\Phi_{\ell,:}^{(h,o)}\|_1 \right) \quad (\text{h,o})$$

$$+ \|\Phi^{(h,h)}\|_{\text{F}}^2 - 2 \left\langle \mathbf{B}_t^{(h,h)}, \Phi^{(h,h)} \right\rangle_{\text{F}} + \frac{\lambda_t}{\Gamma_t} \|\Phi^{(h,h)}\|_{1,1} \quad (\text{h,h})$$

$$= f_t^{(o,o)}(\Phi^{(o,o)}) + \sum_{k=1}^{N-d_y} f_{t,k}^{(o,h)}(\Phi_{:,k}^{(o,h)}) + \sum_{\ell=1}^{M-d_x} f_{t,\ell}^{(h,o)}((\Phi_{\ell,:}^{(h,o)})^\top) + f_t^{(h,h)}(\Phi^{(h,h)}) \\ = H_t(\Phi).$$

Thus, $\bar{F}_t(\Phi) = H_t(\Phi)$. Combining this identity with $\Gamma_t^{-1} F_t(\Phi) = \bar{F}_t(\Phi) + c_t$ gives

$$\frac{1}{\Gamma_t} F_t(\Phi) = H_t(\Phi) + c_t.$$

Since multiplication by the positive scalar Γ_t and addition of c_t do not change the minimizers, we have

$$\underset{\Phi}{\operatorname{argmin}} F_t(\Phi) = \underset{\Phi}{\operatorname{argmin}} H_t(\Phi).$$

Since H_t is separable across blocks, a global minimizer can be assembled from the blockwise minimizers. Let $\hat{\Phi}_t$ be the blockwise-assembled matrix whose blocks solve the corresponding subproblems:

$$\begin{aligned} \hat{\Phi}_t^{(o,o)} &\in \underset{\Theta \in \mathbb{R}^{d_x \times d_y}}{\operatorname{argmin}} f_t^{(o,o)}(\Theta), \\ \hat{\Phi}_{t, :, k}^{(o,h)} &\in \underset{\beta \in \mathbb{R}^{d_x}}{\operatorname{argmin}} f_{t,k}^{(o,h)}(\beta) \quad \forall k \in [N - d_y], \\ \left(\hat{\Phi}_{t, \ell, :}^{(h,o)} \right)^\top &\in \underset{\gamma \in \mathbb{R}^{d_y}}{\operatorname{argmin}} f_{t,\ell}^{(h,o)}(\gamma) \quad \forall \ell \in [M - d_x], \\ \hat{\Phi}_t^{(h,h)} &\in \underset{\Psi \in \mathbb{R}^{(M-d_x) \times (N-d_y)}}{\operatorname{argmin}} f_t^{(h,h)}(\Psi). \end{aligned}$$

For any $\Phi \in \mathbb{R}^{M \times N}$, blockwise optimality implies

$$\begin{aligned} f_t^{(o,o)}(\hat{\Phi}_t^{(o,o)}) &\leq f_t^{(o,o)}(\Phi^{(o,o)}), \quad f_{t,k}^{(o,h)}(\hat{\Phi}_{t, :, k}^{(o,h)}) \leq f_{t,k}^{(o,h)}(\Phi_{:,k}^{(o,h)}), \\ f_{t,\ell}^{(h,o)}((\hat{\Phi}_{t, \ell, :}^{(h,o)})^\top) &\leq f_{t,\ell}^{(h,o)}((\Phi_{\ell, :}^{(h,o)})^\top), \quad f_t^{(h,h)}(\hat{\Phi}_t^{(h,h)}) \leq f_t^{(h,h)}(\Phi^{(h,h)}). \end{aligned}$$

Summing these inequalities gives $H_t(\hat{\Phi}_t) \leq H_t(\Phi)$ for all Φ . Therefore, $\hat{\Phi}_t$ is a global minimizer of H_t (and hence of F_t). Conversely, any global minimizer must minimize each blockwise objective. Otherwise, replacing a non-optimal block by its blockwise minimizer would strictly decrease H_t , a contradiction. Therefore, we can conclude that $\hat{\Phi}_t$ can be computed by solving these subproblems independently. $\square$

C. Proof of Theorem 6.1

C.1. Technical Lemmas for the DR Bilinear Lasso Estimator

Lemma C.1 (Error bound for the Bilinear Lasso estimator). *Let $\{\mathbf{x}_\tau\}_{\tau \in [t]} \subset [-1, 1]^{d_1}$ and $\{\mathbf{y}_\tau\}_{\tau \in [t]} \subset [-1, 1]^{d_2}$ be covariates, and let*

$$r_\tau = \mathbf{x}_\tau^\top \bar{\Phi} \mathbf{y}_\tau + e_\tau$$

for some $\bar{\Phi} \in \mathbb{R}^{d_1 \times d_2}$ and $e_\tau \in \mathbb{R}$. For $\lambda > 0$, let

$$\hat{\Phi}_t = \underset{\Phi}{\operatorname{argmin}} \sum_{\tau=1}^t (r_\tau - \mathbf{x}_\tau^\top \Phi \mathbf{y}_\tau)^2 + \lambda \|\Phi\|_{1,1}.$$

Let $\bar{\mathcal{S}} := \{(i, j) \in [d_1] \times [d_2] : \bar{\Phi}_{ij} \neq 0\}$, $\mathbf{z}_\tau = \operatorname{vec}(\mathbf{x}_\tau \mathbf{y}_\tau^\top) \in \mathbb{R}^{d_1 d_2}$, $\Sigma_t = \sum_{\tau=1}^t \mathbf{z}_\tau \mathbf{z}_\tau^\top$. Suppose $\lambda_{\min}(\Sigma_t) > 0$ and

$$\left\| \sum_{\tau=1}^t e_\tau \mathbf{x}_\tau \mathbf{y}_\tau^\top \right\|_{\infty, \infty} \leq \lambda/2.$$

Then,

$$\sum_{\tau=1}^t \left(\mathbf{x}_\tau^\top (\hat{\Phi}_t - \bar{\Phi}) \mathbf{y}_\tau \right)^2 \leq \frac{4\lambda^2 |\bar{\mathcal{S}}|}{\lambda_{\min}(\Sigma_t)}.$$

Thus, we have

$$\left\| \operatorname{vec}(\hat{\Phi}_t - \bar{\Phi}) \right\|_{\Sigma_t} \leq 2\lambda \sqrt{\frac{|\bar{\mathcal{S}}|}{\lambda_{\min}(\Sigma_t)}}.$$

Proof. By the definition of $\hat{\Phi}_t$, we have

$$\sum_{\tau=1}^t (\mathbf{x}_\tau^\top (\bar{\Phi} - \hat{\Phi}_t) \mathbf{y}_\tau + e_\tau)^2 + \lambda \|\hat{\Phi}_t\|_{1,1} \leq \sum_{\tau=1}^t e_\tau^2 + \lambda \|\bar{\Phi}\|_{1,1},$$

which implies

$$\sum_{\tau=1}^t (\mathbf{x}_\tau^\top (\hat{\Phi}_t - \bar{\Phi}) \mathbf{y}_\tau)^2 + \lambda \|\hat{\Phi}_t\|_{1,1} \leq 2 \sum_{\tau=1}^t \mathbf{x}_\tau^\top (\hat{\Phi}_t - \bar{\Phi}) \mathbf{y}_\tau e_\tau + \lambda \|\bar{\Phi}\|_{1,1}. \quad (13)$$

Meanwhile, for any matrix $\mathbf{M} \in \mathbb{R}^{d_1 \times d_2}$, and vectors $\mathbf{a} \in \mathbb{R}^{d_1}$, $\mathbf{b} \in \mathbb{R}^{d_2}$, we have

$$|\mathbf{a}^\top \mathbf{M} \mathbf{b}| = |\langle \mathbf{M}, \mathbf{a} \mathbf{b}^\top \rangle_F| \leq \|\mathbf{M}\|_{1,1} \|\mathbf{a} \mathbf{b}^\top\|_{\infty, \infty}.$$

Using this inequality, we obtain

$$\begin{aligned} 2 \sum_{\tau=1}^t \mathbf{x}_\tau^\top (\hat{\Phi}_t - \bar{\Phi}) \mathbf{y}_\tau e_\tau &= 2 \langle \hat{\Phi}_t - \bar{\Phi}, \sum_{\tau=1}^t e_\tau \mathbf{x}_\tau \mathbf{y}_\tau^\top \rangle_F \\ &\leq 2 \|\hat{\Phi}_t - \bar{\Phi}\|_{1,1} \left\| \sum_{\tau=1}^t e_\tau \mathbf{x}_\tau \mathbf{y}_\tau^\top \right\|_{\infty, \infty} \\ &\leq \lambda \|\hat{\Phi}_t - \bar{\Phi}\|_{1,1} \\ &= \lambda \sum_{(i,j) \in \bar{\mathcal{S}}} |\hat{\Phi}_t(i,j) - \bar{\Phi}(i,j)| + \lambda \sum_{(i,j) \notin \bar{\mathcal{S}}} |\hat{\Phi}_t(i,j)|. \end{aligned} \quad (14)$$

On the other hand, we also have

$$\begin{aligned} \|\hat{\Phi}_t\|_{1,1} &= \sum_{(i,j) \in \bar{\mathcal{S}}} |\hat{\Phi}_t(i,j)| + \sum_{(i,j) \notin \bar{\mathcal{S}}} |\hat{\Phi}_t(i,j)| \\ &\geq \sum_{(i,j) \in \bar{\mathcal{S}}} |\bar{\Phi}(i,j)| - \sum_{(i,j) \in \bar{\mathcal{S}}} |\hat{\Phi}_t(i,j) - \bar{\Phi}(i,j)| + \sum_{(i,j) \notin \bar{\mathcal{S}}} |\hat{\Phi}_t(i,j)| \\ &= \|\bar{\Phi}\|_{1,1} - \sum_{(i,j) \in \bar{\mathcal{S}}} |\hat{\Phi}_t(i,j) - \bar{\Phi}(i,j)| + \sum_{(i,j) \notin \bar{\mathcal{S}}} |\hat{\Phi}_t(i,j)|. \end{aligned} \quad (15)$$

Substituting the upper bound from (14) and the lower bound from (15) into (13), we obtain

$$\sum_{\tau=1}^t (\mathbf{x}_\tau^\top (\hat{\Phi}_t - \bar{\Phi}) \mathbf{y}_\tau)^2 \leq 2\lambda \sum_{(i,j) \in \bar{\mathcal{S}}} |\hat{\Phi}_t(i,j) - \bar{\Phi}(i,j)|.$$

Now consider $\sum_{\tau=1}^t (\mathbf{x}_\tau^\top (\hat{\Phi}_t - \bar{\Phi}) \mathbf{y}_\tau)^2$.

$$\begin{aligned} \sum_{\tau=1}^t (\mathbf{x}_\tau^\top (\hat{\Phi}_t - \bar{\Phi}) \mathbf{y}_\tau)^2 &= \sum_{\tau=1}^t \mathbf{y}_\tau^\top (\hat{\Phi}_t - \bar{\Phi})^\top \mathbf{x}_\tau \mathbf{x}_\tau^\top (\hat{\Phi}_t - \bar{\Phi}) \mathbf{y}_\tau \\ &= \sum_{\tau=1}^t \text{vec}(\hat{\Phi}_t - \bar{\Phi})^\top \mathbf{z}_\tau \mathbf{z}_\tau^\top \text{vec}(\hat{\Phi}_t - \bar{\Phi}) \\ &\geq \lambda_{\min}(\Sigma_t) \sum_{(i,j) \in \bar{\mathcal{S}}} |\hat{\Phi}_t(i,j) - \bar{\Phi}(i,j)|^2 \\ &\geq \frac{\lambda_{\min}(\Sigma_t)}{|\bar{\mathcal{S}}|} \left(\sum_{(i,j) \in \bar{\mathcal{S}}} |\hat{\Phi}_t(i,j) - \bar{\Phi}(i,j)| \right)^2 \end{aligned}$$

Thus

$$\begin{aligned} \sum_{\tau=1}^t (\mathbf{x}_\tau^\top (\hat{\Phi}_t - \bar{\Phi}) \mathbf{y}_\tau)^2 &\leq 2\lambda \sum_{(i,j) \in \bar{\mathcal{S}}} |\hat{\Phi}_t(i,j) - \bar{\Phi}(i,j)| \\ &\leq 2\lambda \sqrt{\frac{|\bar{\mathcal{S}}|}{\lambda_{\min}(\Sigma_t)}} \sqrt{\sum_{\tau=1}^t (\mathbf{x}_\tau^\top (\hat{\Phi}_t - \bar{\Phi}) \mathbf{y}_\tau)^2} \\ &\leq \frac{2\lambda^2 |\bar{\mathcal{S}}|}{\lambda_{\min}(\Sigma_t)} + \frac{1}{2} \sum_{\tau=1}^t (\mathbf{x}_\tau^\top (\hat{\Phi}_t - \bar{\Phi}) \mathbf{y}_\tau)^2 \end{aligned}$$

Rearranging the terms, we have

$$\sum_{\tau=1}^t (\mathbf{x}_\tau^\top (\hat{\Phi}_t - \bar{\Phi}) \mathbf{y}_\tau)^2 \leq \frac{4\lambda^2 |\bar{\mathcal{S}}|}{\lambda_{\min}(\Sigma_t)}.$$

The second claim follows because

$$\left\| \text{vec}(\hat{\Phi}_t - \bar{\Phi}) \right\|_{\Sigma_t}^2 = \sum_{\tau=1}^t (\mathbf{x}_\tau^\top (\hat{\Phi}_t - \bar{\Phi}) \mathbf{y}_\tau)^2.$$

□

The Freedman-based argument below sharpens the exploration-phase bound. We use this concentration refinement as an analysis tool rather than as the bilinear-specific contribution of the paper. The bilinear-specific ingredients are the effective matrix parameterization, the DR-Lasso analysis, and the exact blockwise solver.

Lemma C.2 (Matrix concentration). *Fix $p \in (1/2, 1)$ and $\delta \in (0, 1)$. For each round τ , let $\tilde{a}_\tau \in [K]$ be the pseudo-action with known probabilities $\{\phi_{a,\tau}\}_{a \in [K]}$, i.e.,*

$$\mathbb{P}(\tilde{a}_\tau = a \mid \mathcal{F}_{\tau-1}) = \phi_{a,\tau} \quad \forall a \in [K]. \quad (16)$$

Let $\{\tilde{\mathbf{x}}_a\}_{a \in [K]} \subset \mathbb{R}^K$ denote the augmented feature vectors, and assume that

$$\mathbf{V} := \sum_{a \in [K]} \tilde{\mathbf{x}}_a \tilde{\mathbf{x}}_a^\top$$

is positive definite. Define

$$\mathbf{V}_t := \sum_{\tau=1}^t \sum_{a \in [K]} \tilde{\mathbf{x}}_a \tilde{\mathbf{x}}_a^\top,$$

$$\mathbf{A}_t := \sum_{\tau=1}^t \sum_{a \in [K]} \frac{\mathbb{1}(\tilde{a}_\tau = a)}{\phi_{a,\tau}} \tilde{\mathbf{x}}_a \tilde{\mathbf{x}}_a^\top.$$

Let

$$\phi_{\min} := \min_{\tau \geq 1} \min_{a \in [K]} \phi_{a,\tau} = \frac{1-p}{K-1},$$

$$b := \phi_{\min}^{-1} - 1 = \frac{K-2+p}{1-p}.$$

Then for any $\epsilon \in (0, 1)$ and any $t \geq 1$ satisfying

$$t \geq \frac{8b}{3\epsilon^2} \log \left(\frac{2Kt^2}{\delta} \right), \quad (17)$$

we have, with probability at least $1 - \delta/t^2$,

$$\left\| \mathbf{I}_K - \mathbf{V}_t^{-1/2} \mathbf{A}_t \mathbf{V}_t^{-1/2} \right\|_2 \leq \epsilon. \quad (18)$$

Proof. Since the augmented features are fixed over time, define the deterministic matrix

$$\mathbf{V} := \sum_{a \in [K]} \tilde{\mathbf{x}}_a \tilde{\mathbf{x}}_a^\top,$$

so that $\mathbf{V}_t = t\mathbf{V}$. Let $\tilde{\mathbf{X}} \in \mathbb{R}^{K \times K}$ be the matrix whose columns are the augmented features:

$$\tilde{\mathbf{X}} := [\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_K],$$

so $\mathbf{V} = \tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top$. Since $\lambda_{\min}(\mathbf{V}) > 0$, $\mathbf{V}$ is positive definite. Since $\tilde{\mathbf{X}}$ is square and $\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top \succ 0$, it follows that $\tilde{\mathbf{X}}$ is invertible. Define normalized vectors and the associated square matrix

$$\mathbf{v}_a := \mathbf{V}^{-1/2} \tilde{\mathbf{x}}_a \in \mathbb{R}^K, \quad a \in [K],$$

$$\mathbf{Q} := [\mathbf{v}_1, \dots, \mathbf{v}_K] = \mathbf{V}^{-1/2} \tilde{\mathbf{X}}.$$

Then

$$\mathbf{Q}\mathbf{Q}^\top = \mathbf{V}^{-1/2} \tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top \mathbf{V}^{-1/2} = \mathbf{V}^{-1/2} \mathbf{V} \mathbf{V}^{-1/2} = \mathbf{I}_K. \quad (19)$$

Since $\mathbf{Q}$ is square, (19) implies that $\mathbf{Q}$ is orthogonal, and therefore

$$\mathbf{v}_a^\top \mathbf{v}_{a'} = \mathbb{1}(a = a') \quad \forall a, a' \in [K].$$

Next, compute the normalized Gram estimator:

$$\begin{aligned} \mathbf{V}_t^{-1/2} \mathbf{A}_t \mathbf{V}_t^{-1/2} &= (t\mathbf{V})^{-1/2} \left(\sum_{\tau=1}^t \sum_{a \in [K]} \frac{\mathbb{1}(\tilde{a}_\tau = a)}{\phi_{a,\tau}} \tilde{\mathbf{x}}_a \tilde{\mathbf{x}}_a^\top \right) (t\mathbf{V})^{-1/2} \\ &= \frac{1}{t} \sum_{\tau=1}^t \sum_{a \in [K]} \frac{\mathbb{1}(\tilde{a}_\tau = a)}{\phi_{a,\tau}} \left(\mathbf{V}^{-1/2} \tilde{\mathbf{x}}_a \right) \left(\mathbf{V}^{-1/2} \tilde{\mathbf{x}}_a \right)^\top \\ &= \sum_{a \in [K]} \hat{s}_{t,a} \mathbf{v}_a \mathbf{v}_a^\top, \end{aligned}$$

where

$$\hat{s}_{t,a} := \frac{1}{t} \sum_{\tau=1}^t \frac{\mathbb{1}(\tilde{a}_\tau = a)}{\phi_{a,\tau}}.$$

Since $\mathbf{I}_K = \mathbf{Q}\mathbf{Q}^\top = \sum_{a \in [K]} \mathbf{v}_a \mathbf{v}_a^\top$, we obtain

$$\mathbf{I}_K - \mathbf{V}_t^{-1/2} \mathbf{A}_t \mathbf{V}_t^{-1/2} = \sum_{a \in [K]} (1 - \hat{s}_{t,a}) \mathbf{v}_a \mathbf{v}_a^\top. \quad (20)$$

Since $\{\mathbf{v}_a\}_{a \in [K]}$ is an orthonormal basis, the matrix in (20) is diagonal in this basis, with eigenvalues $\{1 - \hat{s}_{t,a}\}_{a \in [K]}$. Hence

$$\begin{aligned} \left\| \mathbf{I}_K - \mathbf{V}_t^{-1/2} \mathbf{A}_t \mathbf{V}_t^{-1/2} \right\|_2 &= \max_{a \in [K]} |1 - \hat{s}_{t,a}| \\ &= \max_{a \in [K]} \left| \frac{1}{t} \sum_{\tau=1}^t \left(\frac{\mathbb{1}(\tilde{a}_\tau = a)}{\phi_{a,\tau}} - 1 \right) \right|. \end{aligned} \quad (21)$$

Fix an arm $a \in [K]$. Define the martingale difference sequence

$$Z_{\tau,a} := \frac{\mathbb{1}(\tilde{a}_\tau = a)}{\phi_{a,\tau}} - 1, \quad S_{t,a} := \sum_{\tau=1}^t Z_{\tau,a}.$$

By (16),

$$\mathbb{E}[Z_{\tau,a} \mid \mathcal{F}_{\tau-1}] = \frac{1}{\phi_{a,\tau}} \mathbb{E}[\mathbb{1}(\tilde{a}_\tau = a) \mid \mathcal{F}_{\tau-1}] - 1 = \frac{1}{\phi_{a,\tau}} \phi_{a,\tau} - 1 = 0,$$

so $\{S_{t,a}\}_{t \geq 0}$ is a scalar martingale adapted to $\{\mathcal{F}_t\}$. For all τ ,

$$Z_{\tau,a} \leq \frac{1}{\phi_{a,\tau}} - 1 \leq \frac{1}{\phi_{\min}} - 1 = b.$$

Conditioned on $\mathcal{F}_{\tau-1}$, $\mathbb{1}(\tilde{a}_\tau = a)$ is Bernoulli with mean $\phi_{a,\tau}$, hence

$$\begin{aligned} \mathbb{E}[Z_{\tau,a}^2 \mid \mathcal{F}_{\tau-1}] &= \phi_{a,\tau} \left(\frac{1}{\phi_{a,\tau}} - 1 \right)^2 + (1 - \phi_{a,\tau})(-1)^2 \\ &= \frac{1}{\phi_{a,\tau}} - 1 \leq \frac{1}{\phi_{\min}} - 1 = b. \end{aligned}$$

Therefore, with $W_{t,a} := \sum_{\tau=1}^t \mathbb{E}[Z_{\tau,a}^2 \mid \mathcal{F}_{\tau-1}]$, we have

$$W_{t,a} \leq tb.$$

Apply Theorem D.2 to the martingale $\{S_{s,a}\}_{s=0}^t$ with $R = b$ and variance proxy $\sigma^2 = tb$. For the upper tail, for any $\epsilon \in (0, 1)$,

$$\mathbb{P}(S_{t,a} \geq t\epsilon) \leq \exp\left(-\frac{(t\epsilon)^2/2}{tb + b(t\epsilon)/3}\right).$$

Since $\epsilon \in (0, 1)$ implies $tb + b(t\epsilon)/3 \leq (4/3)tb$, we obtain

$$\mathbb{P}(S_{t,a} \geq t\epsilon) \leq \exp\left(-\frac{3t\epsilon^2}{8b}\right). \quad (22)$$

The same argument applies to $-S_{t,a}$ since $-Z_{\tau,a} \leq 1 \leq b$:

$$\mathbb{P}(S_{t,a} \leq -t\epsilon) \leq \exp\left(-\frac{3t\epsilon^2}{8b}\right). \quad (23)$$

Combining (22) and (23),

$$\mathbb{P}(|S_{t,a}| \geq t\epsilon) \leq 2 \exp\left(-\frac{3t\epsilon^2}{8b}\right).$$

By a union bound over arms,

$$\mathbb{P}\left(\max_{a \in [K]} |S_{t,a}| \geq t\epsilon\right) \leq 2K \exp\left(-\frac{3t\epsilon^2}{8b}\right). \quad (24)$$

If t satisfies (17), then the right-hand side of (24) is at most δ/t^2 . Equivalently, with probability at least $1 - \delta/t^2$,

$$\max_{a \in [K]} \left| \frac{1}{t} S_{t,a} \right| \leq \epsilon. \quad (25)$$

Combining (25) with (21) yields (18). $\square$

C.2. Consistency of the DR Bilinear Lasso estimator

Lemma C.3 (Consistency of the DR Bilinear Lasso estimator). *Let d_{h_x} and d_{h_y} denote the effective dimensions associated with the orthogonal complements of the observable feature subspaces, as defined in Section 4.2. For any t satisfying*

$$t \geq \frac{512\tilde{\sigma}_{\max,z}^2 MN(MN - 2 + p)}{3\tilde{\sigma}_{\min,z}^2(1 - p)} \log\left(\frac{2MNt^2}{\delta}\right),$$

the following inequality holds with probability at least $1 - 3\delta$:

$$\max_{(i,j) \in \mathcal{A}} |\tilde{\mathbf{x}}_i^\top (\hat{\Phi}_t - \Phi_*) \tilde{\mathbf{y}}_j| \leq \frac{8\sigma\kappa_{\max,x}\kappa_{\max,y}}{\tilde{\sigma}_{\min,z}p} \sqrt{\frac{2(d_x + d_{h_x})(d_y + d_{h_y})}{t} \log\left(\frac{2MNt^2}{\delta}\right)}$$

Proof. To apply Lemma C.1, we verify its noise condition for the DR pseudo-reward regression. It suffices to upper bound

$$\left\| \sum_{\tau=1}^t \sum_{(i,j) \in \mathcal{A}} (\tilde{r}_{(i,j),\tau} - \tilde{\mathbf{x}}_i^\top \Phi_* \tilde{\mathbf{y}}_j) \tilde{\mathbf{x}}_i \tilde{\mathbf{y}}_j^\top \right\|_{\infty, \infty}.$$

We decompose the main term as follows:

$$\begin{aligned} & \left\| \sum_{\tau=1}^t \sum_{(i,j) \in \mathcal{A}} (\tilde{r}_{(i,j),\tau} - \tilde{\mathbf{x}}_i^\top \Phi_* \tilde{\mathbf{y}}_j) \tilde{\mathbf{x}}_i \tilde{\mathbf{y}}_j^\top \right\|_{\infty, \infty} \\ &= \left\| \sum_{\tau=1}^t \sum_{(i,j) \in \mathcal{A}} \left(1 - \frac{\mathbf{1}(\tilde{a}_\tau = (i,j))}{\phi_{(i,j),\tau}}\right) \tilde{\mathbf{x}}_i \tilde{\mathbf{x}}_i^\top (\hat{\Phi}_t - \Phi_*) \tilde{\mathbf{y}}_j \tilde{\mathbf{y}}_j^\top \right. \\ & \quad \left. + \frac{\mathbf{1}(\tilde{a}_\tau = (i,j))}{\phi_{(i,j),\tau}} (\tilde{r}_{(i,j),\tau} - \tilde{\mathbf{x}}_i^\top \Phi_* \tilde{\mathbf{y}}_j) \tilde{\mathbf{x}}_i \tilde{\mathbf{y}}_j^\top \right\|_{\infty, \infty} \\ &\leq \underbrace{\left\| \sum_{\tau=1}^t \sum_{(i,j) \in \mathcal{A}} \left(1 - \frac{\mathbf{1}(\tilde{a}_\tau = (i,j))}{\phi_{(i,j),\tau}}\right) \tilde{\mathbf{x}}_i \tilde{\mathbf{x}}_i^\top (\hat{\Phi}_t - \Phi_*) \tilde{\mathbf{y}}_j \tilde{\mathbf{y}}_j^\top \right\|_{\infty, \infty}}_{(i)} \\ & \quad + \underbrace{\left\| \sum_{\tau=1}^t \sum_{(i,j) \in \mathcal{A}} \frac{\mathbf{1}(\tilde{a}_\tau = (i,j))}{\phi_{(i,j),\tau}} (\tilde{r}_{(i,j),\tau} - \tilde{\mathbf{x}}_i^\top \Phi_* \tilde{\mathbf{y}}_j) \tilde{\mathbf{x}}_i \tilde{\mathbf{y}}_j^\top \right\|_{\infty, \infty}}_{(ii)}. \end{aligned}$$

Let $\mathcal{M}_t$ denote the event that the pseudo-action $\tilde{a}_t$ matches the chosen action a_t within a specified number of resampling attempts. Then, with probability at least $1 - \delta$, the event $\mathcal{M}_\tau$ occurs for all $\tau \leq t$, so the pseudo-action and the chosen action match at each round up to t . Thus, term (ii) is equal to

$$\begin{aligned}
 & \left\| \sum_{\tau=1}^t \sum_{(i,j) \in \mathcal{A}} \frac{\mathbb{1}(\tilde{a}_\tau = (i,j))}{\phi_{(i,j),\tau}} (r_{(i,j),\tau} - \tilde{\mathbf{x}}_i^\top \Phi_* \tilde{\mathbf{y}}_j) \tilde{\mathbf{x}}_i \tilde{\mathbf{y}}_j^\top \right\|_{\infty, \infty} \\
 &= \left\| \sum_{\tau=1}^t \sum_{(i,j) \in \mathcal{A}} \frac{\mathbb{1}(a_\tau = (i,j))}{p} (r_{(i,j),\tau} - \tilde{\mathbf{x}}_i^\top \Phi_* \tilde{\mathbf{y}}_j) \tilde{\mathbf{x}}_i \tilde{\mathbf{y}}_j^\top \right\|_{\infty, \infty} \\
 &= \frac{1}{p} \left\| \sum_{\tau=1}^t \epsilon_\tau \tilde{\mathbf{x}}_{a_{\tau,1}} \tilde{\mathbf{y}}_{a_{\tau,2}}^\top \right\|_{\infty, \infty} \\
 &= \frac{1}{p} \max_{(i,j) \in \mathcal{A}} \left| \sum_{\tau=1}^t \epsilon_\tau ((\mathbf{e}_\mathbf{x})_i^\top \tilde{\mathbf{x}}_{i_\tau}) ((\mathbf{e}_\mathbf{y})_j^\top \tilde{\mathbf{y}}_{j_\tau}) \right|.
 \end{aligned}$$

For each fixed $(i,j) \in \mathcal{A}$, define

$$M_{t,i,j} := \sum_{\tau=1}^t \epsilon_\tau ((\mathbf{e}_\mathbf{x})_i^\top \tilde{\mathbf{x}}_{i_\tau}) ((\mathbf{e}_\mathbf{y})_j^\top \tilde{\mathbf{y}}_{j_\tau}).$$

Then $\{M_{s,i,j}\}_{s=0}^t$ is a martingale with respect to $\{\mathcal{F}_s\}_{s=0}^t$. By Lemma D.1, with probability at least $1 - \delta/t^2$, we have

$$\begin{aligned}
 & \max_{(i,j) \in \mathcal{A}} \left| \sum_{\tau=1}^t \epsilon_\tau ((\mathbf{e}_\mathbf{x})_i^\top \tilde{\mathbf{x}}_{a_{\tau,1}}) ((\mathbf{e}_\mathbf{y})_j^\top \tilde{\mathbf{y}}_{a_{\tau,2}}) \right| \\
 &\leq \max_{(i,j) \in \mathcal{A}} \sigma \sqrt{2 \sum_{\tau=1}^t ((\mathbf{e}_\mathbf{x})_i^\top \tilde{\mathbf{x}}_{a_{\tau,1}})^2 ((\mathbf{e}_\mathbf{y})_j^\top \tilde{\mathbf{y}}_{a_{\tau,2}})^2 \log \frac{2MNt^2}{\delta}} \\
 &\leq \max_{(i,j) \in \mathcal{A}} \sigma \sqrt{2 \sqrt{\sum_{\tau=1}^t [(\mathbf{e}_\mathbf{x})_i^\top \tilde{\mathbf{x}}_{a_{\tau,1}}]^4} \sqrt{\sum_{\tau=1}^t [(\mathbf{e}_\mathbf{y})_j^\top \tilde{\mathbf{y}}_{a_{\tau,2}}]^4} \log \frac{2MNt^2}{\delta}} \\
 &\leq \sigma \kappa_{\max,x} \kappa_{\max,y} \sqrt{2t \log \frac{2MNt^2}{\delta}}
 \end{aligned}$$

where $\kappa_{\max,x}^4 = \max_{i \in [M]} \max_{i' \in [M]} [(\mathbf{e}_\mathbf{x})_i^\top \tilde{\mathbf{x}}_{i'}]^4$ and $\kappa_{\max,y}^4 = \max_{j \in [N]} \max_{j' \in [N]} [(\mathbf{e}_\mathbf{y})_j^\top \tilde{\mathbf{y}}_{j'}]^4$. Thus, we have

$$\left\| \sum_{\tau=1}^t \sum_{(i,j) \in \mathcal{A}} \frac{\mathbb{1}(\tilde{a}_\tau = (i,j))}{\phi_{(i,j),\tau}} (r_{(i,j),\tau} - \tilde{\mathbf{x}}_i^\top \Phi_* \tilde{\mathbf{y}}_j) \tilde{\mathbf{x}}_i \tilde{\mathbf{y}}_j^\top \right\|_{\infty, \infty} \leq \frac{\sigma \kappa_{\max,x} \kappa_{\max,y}}{p} \sqrt{2t \log \frac{2MNt^2}{\delta}}.$$

We next bound term (i). Specifically, we define $\tilde{\mathbf{z}}_{(i,j)} := \text{vec}(\tilde{\mathbf{x}}_i \tilde{\mathbf{y}}_j^\top)$, which represents the vectorization of the outer product between the user and item feature vectors. Define the following Gram matrices:

$$\begin{aligned}
 \mathbf{A}_t &:= \sum_{\tau=1}^t \sum_{(i,j) \in \mathcal{A}} \frac{\mathbb{1}(\tilde{a}_\tau = (i,j))}{\phi_{(i,j),\tau}} \tilde{\mathbf{z}}_{(i,j)} \tilde{\mathbf{z}}_{(i,j)}^\top, \\
 \mathbf{V}_t &:= \sum_{\tau=1}^t \sum_{(i,j) \in \mathcal{A}} \tilde{\mathbf{z}}_{(i,j)} \tilde{\mathbf{z}}_{(i,j)}^\top, \\
 \Sigma_t &:= \sum_{\tau=1}^t \tilde{\mathbf{z}}_{a_\tau} \tilde{\mathbf{z}}_{a_\tau}^\top.
 \end{aligned}$$

On the event that $\mathcal{M}_\tau$ holds for all $\tau \leq t$, the matrix $\mathbf{A}_t$ satisfies

$$\mathbf{A}_t = \sum_{\tau=1}^t \frac{1}{p} \tilde{\mathbf{z}}_{a_\tau} \tilde{\mathbf{z}}_{a_\tau}^\top = \frac{1}{p} \boldsymbol{\Sigma}_t.$$

Then the following holds:

$$\begin{aligned} & \left\| \sum_{\tau=1}^t \sum_{(i,j) \in \mathcal{A}} \left(1 - \frac{\mathbb{1}(\tilde{a}_\tau = (i,j))}{\phi_{(i,j),\tau}} \right) \tilde{\mathbf{x}}_i \tilde{\mathbf{x}}_i^\top (\check{\boldsymbol{\Phi}}_t - \boldsymbol{\Phi}_*) \tilde{\mathbf{y}}_j \tilde{\mathbf{y}}_j^\top \right\|_{\infty, \infty} \\ &= \left\| \sum_{\tau=1}^t \sum_{(i,j) \in \mathcal{A}} \left(1 - \frac{\mathbb{1}(\tilde{a}_\tau = (i,j))}{\phi_{(i,j),\tau}} \right) \tilde{\mathbf{z}}_{(i,j)} \tilde{\mathbf{z}}_{(i,j)}^\top [\text{vec}(\check{\boldsymbol{\Phi}}_t) - \text{vec}(\boldsymbol{\Phi}_*)] \right\|_\infty \\ &= \|(\mathbf{V}_t - \mathbf{A}_t)(\text{vec}(\check{\boldsymbol{\Phi}}_t) - \text{vec}(\boldsymbol{\Phi}_*))\|_\infty \\ &= \max_{k \in [MN]} |(\mathbf{e}_z)_k^\top (\mathbf{V}_t - \mathbf{A}_t)(\text{vec}(\check{\boldsymbol{\Phi}}_t) - \text{vec}(\boldsymbol{\Phi}_*))| \\ &\leq \max_{k \in [MN]} \left\| (\mathbf{e}_z)_k^\top (\mathbf{V}_t - \mathbf{A}_t) \mathbf{A}_t^{-\frac{1}{2}} \right\|_2 \left\| \text{vec}(\check{\boldsymbol{\Phi}}_t) - \text{vec}(\boldsymbol{\Phi}_*) \right\|_{\mathbf{A}_t}. \end{aligned}$$

Since $\check{\boldsymbol{\Phi}}_t$ is the minimizer of (2), by applying Lemma C.1, we obtain

$$\begin{aligned} \left\| \text{vec}(\check{\boldsymbol{\Phi}}_t) - \text{vec}(\boldsymbol{\Phi}_*) \right\|_{\mathbf{A}_t} &= \sqrt{\frac{1}{p}} \left\| \text{vec}(\check{\boldsymbol{\Phi}}_t) - \text{vec}(\boldsymbol{\Phi}_*) \right\|_{\boldsymbol{\Sigma}_t} \\ &\leq \sqrt{\frac{1}{p}} \cdot 2 \left(2\sigma \kappa_{\max,x} \kappa_{\max,y} \sqrt{2t \log \left(\frac{2MNt^2}{\delta} \right)} \right) \sqrt{\frac{(d_x + d_{h_x})(d_y + d_{h_y})}{\lambda_{\min}(\boldsymbol{\Sigma}_t)}} \\ &= \frac{4\sigma \kappa_{\max,x} \kappa_{\max,y} \sqrt{2t(d_x + d_{h_x})(d_y + d_{h_y}) \log \left(\frac{2MNt^2}{\delta} \right)}}{p \sqrt{\lambda_{\min}(\mathbf{A}_t)}}. \end{aligned}$$

Here, d_{h_x} and d_{h_y} denote the numbers of non-zero rows and columns, respectively, in the unobservable part of $\boldsymbol{\Phi}_*$.

By Lemma C.2, for any $\epsilon \in (0, 1)$ and for all $t \geq \frac{8(MN-2+p)}{3\epsilon^2(1-p)} \log \left(\frac{2MNt^2}{\delta} \right)$, the following inequality holds:

$$\left\| \mathbf{I}_{MN} - \mathbf{V}_t^{-1/2} \mathbf{A}_t \mathbf{V}_t^{-1/2} \right\|_2 \leq \epsilon,$$

which implies $(1 - \epsilon) \mathbf{I}_{MN} \preceq \mathbf{V}_t^{-1/2} \mathbf{A}_t \mathbf{V}_t^{-1/2}$ and $(1 - \epsilon) \mathbf{V}_t \preceq \mathbf{A}_t$. Define $\tilde{\sigma}_{\min,z}^2 = \lambda_{\min}(\sum_{(i,j) \in [M] \times [N]} \tilde{\mathbf{z}}_{(i,j)} \tilde{\mathbf{z}}_{(i,j)}^\top) = \frac{\lambda_{\min}(\mathbf{V}_t)}{t}$. Then,

$$\begin{aligned} \left\| \text{vec}(\check{\boldsymbol{\Phi}}_t) - \text{vec}(\boldsymbol{\Phi}_*) \right\|_{\mathbf{A}_t} &\leq \frac{4\sigma \kappa_{\max,x} \kappa_{\max,y}}{p} \sqrt{\frac{2t(d_x + d_{h_x})(d_y + d_{h_y}) \log \frac{2MNt^2}{\delta}}{(1 - \epsilon) \lambda_{\min}(\mathbf{V}_t)}} \\ &= \frac{4\sigma \kappa_{\max,x} \kappa_{\max,y}}{p \tilde{\sigma}_{\min,z}} \sqrt{\frac{2(d_x + d_{h_x})(d_y + d_{h_y}) \log \frac{2MNt^2}{\delta}}{(1 - \epsilon)}}. \end{aligned}$$

Let

$$\tilde{\sigma}_{\max,z}^2 := \max_{k \in [MN]} (\mathbf{e}_z)_k^\top \sum_{(i,j) \in \mathcal{A}} \tilde{\mathbf{z}}_{(i,j)} \tilde{\mathbf{z}}_{(i,j)}^\top (\mathbf{e}_z)_k.$$

Then, on the event of Lemma C.2, we have

$$\begin{aligned} & \max_{k \in [MN]} \left\| (\mathbf{e}_z)_k^\top (\mathbf{V}_t - \mathbf{A}_t) \mathbf{A}_t^{-1/2} \right\|_2 \\ &= \max_{k \in [MN]} \left\| (\mathbf{e}_z)_k^\top \mathbf{V}_t^{1/2} \left[\mathbf{V}_t^{-1/2} (\mathbf{V}_t - \mathbf{A}_t) \mathbf{V}_t^{-1/2} \right] \mathbf{V}_t^{1/2} \mathbf{A}_t^{-1/2} \right\|_2 \\ &\leq \max_{k \in [MN]} \left\| \mathbf{V}_t^{1/2} (\mathbf{e}_z)_k \right\|_2 \left\| \mathbf{I}_{MN} - \mathbf{V}_t^{-1/2} \mathbf{A}_t \mathbf{V}_t^{-1/2} \right\|_2 \left\| \mathbf{V}_t^{1/2} \mathbf{A}_t^{-1/2} \right\|_2. \end{aligned}$$

Since

$$(1 - \epsilon)\mathbf{V}_t \preceq \mathbf{A}_t,$$

we have

$$\left\| \mathbf{V}_t^{1/2} \mathbf{A}_t^{-1/2} \right\|_2 \leq \frac{1}{\sqrt{1 - \epsilon}}.$$

Also, by the definition of $\tilde{\sigma}_{\max, z}$,

$$\max_{k \in [MN]} \left\| \mathbf{V}_t^{1/2} (\mathbf{e}_z)_k \right\|_2 = \tilde{\sigma}_{\max, z} \sqrt{t}.$$

Therefore,

$$\max_{k \in [MN]} \left\| (\mathbf{e}_z)_k^\top (\mathbf{V}_t - \mathbf{A}_t) \mathbf{A}_t^{-1/2} \right\|_2 \leq \frac{\tilde{\sigma}_{\max, z} \sqrt{t\epsilon}}{\sqrt{1 - \epsilon}}.$$

Therefore, term (i) can be bounded as follows:

$$\begin{aligned} & \left\| (\mathbf{V}_t - \mathbf{A}_t) (\text{vec}(\check{\Phi}_t) - \text{vec}(\Phi_*)) \right\|_\infty \\ & \leq \max_{k \in [MN]} \left\| (\mathbf{e}_z)_k^\top (\mathbf{V}_t - \mathbf{A}_t) \mathbf{A}_t^{-1/2} \right\|_2 \left\| \text{vec}(\check{\Phi}_t) - \text{vec}(\Phi_*) \right\|_{\mathbf{A}_t} \\ & \leq \frac{\tilde{\sigma}_{\max, z} \sqrt{t\epsilon}}{\sqrt{1 - \epsilon}} \cdot \frac{4\sigma\kappa_{\max, x}\kappa_{\max, y}}{p\tilde{\sigma}_{\min, z}} \sqrt{\frac{2(d_x + d_{h_x})(d_y + d_{h_y}) \log\left(\frac{2MNt^2}{\delta}\right)}{1 - \epsilon}} \\ & = \frac{4\epsilon\tilde{\sigma}_{\max, z}\kappa_{\max, x}\kappa_{\max, y}}{p(1 - \epsilon)\tilde{\sigma}_{\min, z}} \sqrt{2t(d_x + d_{h_x})(d_y + d_{h_y}) \log\left(\frac{2MNt^2}{\delta}\right)}. \end{aligned}$$

Now, by setting $\epsilon = \frac{\tilde{\sigma}_{\min, z}}{8\tilde{\sigma}_{\max, z}} \sqrt{\frac{1}{MN}}$, we ensure $1 - \epsilon \geq 1/2$, which implies $\epsilon/(1 - \epsilon) \leq \frac{\tilde{\sigma}_{\min, z}}{4\tilde{\sigma}_{\max, z}} \sqrt{\frac{1}{MN}}$. Hence, we obtain

$$\begin{aligned} \left\| (\mathbf{V}_t - \mathbf{A}_t) (\text{vec}(\check{\Phi}_t) - \text{vec}(\Phi_*)) \right\|_\infty & \leq \frac{\sigma\kappa_{\max, x}\kappa_{\max, y}}{p} \sqrt{2t \log \frac{2MNt^2}{\delta} \frac{(d_x + d_{h_x})}{M} \frac{(d_y + d_{h_y})}{N}} \\ & \leq \frac{\sigma\kappa_{\max, x}\kappa_{\max, y}}{p} \sqrt{2t \log \frac{2MNt^2}{\delta}}. \end{aligned}$$

This implies that, for $t \geq \frac{512\tilde{\sigma}_{\max, z}^2 MN(MN - 2 + p)}{3\tilde{\sigma}_{\min, z}^2 (1 - p)} \log\left(\frac{2MNt^2}{\delta}\right)$, we have

$$\left\| \sum_{\tau=1}^t \sum_{(i,j) \in \mathcal{A}} (\tilde{r}_{(i,j), \tau} - \tilde{\mathbf{x}}_i^\top \Phi_* \tilde{\mathbf{y}}_j) \tilde{\mathbf{x}}_i \tilde{\mathbf{y}}_j^\top \right\|_{\infty, \infty} \leq \frac{2\sigma\kappa_{\max, x}\kappa_{\max, y}}{p} \sqrt{2t \log \frac{2MNt^2}{\delta}}.$$

Note that the Bilinear DR Lasso estimator $\hat{\Phi}_t$ is given as follows:

$$\hat{\Phi}_t := \underset{\Phi}{\operatorname{argmin}} \sum_{\tau=1}^t \mathbb{1}(\mathcal{M}_\tau) \sum_{(i,j) \in \mathcal{A}} (\tilde{r}_{(i,j), \tau} - \tilde{\mathbf{x}}_i^\top \Phi \tilde{\mathbf{y}}_j)^2 + \frac{4\sigma\kappa_{\max, x}\kappa_{\max, y}}{p} \sqrt{2t \log \frac{2MNt^2}{\delta}} \|\Phi\|_{1,1}.$$

Then, by Lemma C.1, we obtain the following bound:

$$\begin{aligned} & \left\| \text{vec}(\hat{\Phi}_t) - \text{vec}(\Phi_*) \right\|_{\mathbf{V}_t} \\ & \leq \frac{8\sigma\kappa_{\max, x}\kappa_{\max, y}}{p} \sqrt{2t \log \frac{2MNt^2}{\delta}} \sqrt{\frac{(d_x + d_{h_x})(d_y + d_{h_y})}{\lambda_{\min}(\mathbf{V}_t)}} \\ & \leq \frac{8\sigma\kappa_{\max, x}\kappa_{\max, y}}{\tilde{\sigma}_{\min, z} p} \sqrt{2(d_x + d_{h_x})(d_y + d_{h_y}) \log \frac{2MNt^2}{\delta}}. \end{aligned}$$

Finally, for each $(i, j) \in \mathcal{A}$, let $\tilde{\mathbf{z}}_{(i,j)} := \text{vec}(\tilde{\mathbf{x}}_i \tilde{\mathbf{y}}_j^\top)$. Then

$$\begin{aligned} \left| \tilde{\mathbf{x}}_i^\top (\hat{\Phi}_t - \Phi_*) \tilde{\mathbf{y}}_j \right| &= \left| \tilde{\mathbf{z}}_{(i,j)}^\top \left(\text{vec}(\hat{\Phi}_t) - \text{vec}(\Phi_*) \right) \right| \\ &\leq \left\| \text{vec}(\hat{\Phi}_t) - \text{vec}(\Phi_*) \right\|_{\mathbf{V}_t} \left\| \tilde{\mathbf{z}}_{(i,j)} \right\|_{\mathbf{V}_t^{-1}}. \end{aligned}$$

Since

$$\mathbf{V}_t = t \sum_{(i,j) \in \mathcal{A}} \tilde{\mathbf{z}}_{(i,j)} \tilde{\mathbf{z}}_{(i,j)}^\top,$$

we have, for each $(i, j) \in \mathcal{A}$,

$$\mathbf{V}_t \succeq t \tilde{\mathbf{z}}_{(i,j)} \tilde{\mathbf{z}}_{(i,j)}^\top.$$

Equivalently, the leverage score satisfies

$$\tilde{\mathbf{z}}_{(i,j)}^\top \left(\sum_{(i',j') \in \mathcal{A}} \tilde{\mathbf{z}}_{(i',j')} \tilde{\mathbf{z}}_{(i',j')}^\top \right)^{-1} \tilde{\mathbf{z}}_{(i,j)} \leq 1.$$

Thus,

$$\left\| \tilde{\mathbf{z}}_{(i,j)} \right\|_{\mathbf{V}_t^{-1}} \leq t^{-1/2}.$$

Taking a union bound over the matching event, the martingale concentration event for term (ii), and the matrix concentration event for term (i), all the above bounds hold with probability at least $1 - 3\delta$. This completes the proof. $\square$

C.3. Regret Bound for Algorithm 1

We prove Theorem 6.1.

Theorem 6.1 (restated). *Under the conditions of Theorem 6.1, for any $\delta \in (0, 1)$ and $p \in (1/2, 1)$, with probability at least $1 - 6\delta$, the cumulative regret of Algorithm 1 is bounded by*

$$\begin{aligned} &\text{Regret}(T) \\ &\leq \frac{1024MN(MN - 2 + p)\tilde{\sigma}_{\max,z}^2}{3(1-p)\tilde{\sigma}_{\min,z}^2} \log \left(\frac{2MNT^2}{\delta} \right) \\ &\quad + \frac{32\sigma\kappa_{\max,x}\kappa_{\max,y}}{\tilde{\sigma}_{\min,z}p} \sqrt{2(d_x + d_{h_x})(d_y + d_{h_y})T \log \left(\frac{2MNT^2}{\delta} \right)} + \sqrt{2T \log \left(\frac{2}{\delta} \right)} + 4\sqrt{T} + 2\delta. \end{aligned}$$

Proof. Since we assume the reward satisfies $|\mathbb{E}[r_{(i,j),t} | \mathcal{F}_{t-1}]| = |\tilde{\mathbf{x}}_t^\top \Phi_* \tilde{\mathbf{y}}_j| \leq 1$, the instantaneous regret is at most 2 at each round. The number of rounds in the exploration phase is at most $|\mathcal{E}_T| \leq \frac{512\tilde{\sigma}_{\max,z}^2 MN(MN-2+p)}{3\tilde{\sigma}_{\min,z}^2(1-p)} \log \left(\frac{2MNT^2}{\delta} \right)$. Hence, the cumulative regret satisfies

$$\begin{aligned} \text{Regret}(T) &\leq 2 \times \frac{512MN(MN - 2 + p)\tilde{\sigma}_{\max,z}^2}{3(1-p)\tilde{\sigma}_{\min,z}^2} \log \left(\frac{2MNT^2}{\delta} \right) + \sum_{t \in [T] \setminus \mathcal{E}_T} \mathbb{E}_{t-1}[y_{*,t}] - \mathbb{E}_{t-1}[y_{a_t,t}] \\ &= \frac{1024MN(MN - 2 + p)\tilde{\sigma}_{\max,z}^2}{3(1-p)\tilde{\sigma}_{\min,z}^2} \log \left(\frac{2MNT^2}{\delta} \right) \\ &\quad + \sum_{t \in [T] \setminus \mathcal{E}_T} \mathbb{1}(a_t = \hat{a}_t) (\mathbb{E}_{t-1}[y_{*,t}] - \mathbb{E}_{t-1}[y_{a_t,t}]) \\ &\quad + \sum_{t \in [T] \setminus \mathcal{E}_T} \mathbb{1}(a_t \neq \hat{a}_t) (\mathbb{E}_{t-1}[y_{*,t}] - \mathbb{E}_{t-1}[y_{a_t,t}]). \end{aligned}$$

When $a_t = \hat{a}_t$, we can bound the regret using Lemma C.3 as follows.

$$\begin{aligned}
 \mathbb{E}_{t-1}[y_{*,t}] - \mathbb{E}_{t-1}[y_{a_t,t}] &= \tilde{\mathbf{x}}_{i_*}^\top \Phi_* \tilde{\mathbf{y}}_{j_*} - \tilde{\mathbf{x}}_{i_t}^\top \Phi_* \tilde{\mathbf{y}}_{j_t} \\
 &= \tilde{\mathbf{x}}_{i_*}^\top \Phi_* \tilde{\mathbf{y}}_{j_*} - \tilde{\mathbf{x}}_{i_*}^\top \hat{\Phi}_{t-1} \tilde{\mathbf{y}}_{j_*} + \tilde{\mathbf{x}}_{i_*}^\top \hat{\Phi}_{t-1} \tilde{\mathbf{y}}_{j_*} - \tilde{\mathbf{x}}_{i_t}^\top \hat{\Phi}_{t-1} \tilde{\mathbf{y}}_{j_t} \\
 &\quad + \tilde{\mathbf{x}}_{i_t}^\top \hat{\Phi}_{t-1} \tilde{\mathbf{y}}_{j_t} - \tilde{\mathbf{x}}_{i_t}^\top \Phi_* \tilde{\mathbf{y}}_{j_t} \\
 &\leq 2 \max_{(i,j) \in \mathcal{A}} \left| \tilde{\mathbf{x}}_i^\top (\Phi_* - \hat{\Phi}_{t-1}) \tilde{\mathbf{y}}_j \right| + \tilde{\mathbf{x}}_{i_*}^\top \hat{\Phi}_{t-1} \tilde{\mathbf{y}}_{j_*} - \tilde{\mathbf{x}}_{i_t}^\top \hat{\Phi}_{t-1} \tilde{\mathbf{y}}_{j_t} \\
 &\leq 2 \max_{(i,j) \in \mathcal{A}} \left| \tilde{\mathbf{x}}_i^\top (\Phi_* - \hat{\Phi}_{t-1}) \tilde{\mathbf{y}}_j \right| \\
 &\leq \frac{16\sigma\kappa_{\max,x}\kappa_{\max,y}}{\tilde{\sigma}_{\min,z}p} \sqrt{\frac{2(d_x + d_{h_x})(d_y + d_{h_y})}{t-1} \log\left(\frac{2MNT^2}{\delta}\right)}.
 \end{aligned}$$

The second inequality uses that $\hat{a}_t$ maximizes the predicted reward under $\hat{\Phi}_{t-1}$. Applying Lemma C.3 with horizon T , the above consistency bound holds simultaneously for all exploitation rounds with probability at least $1 - 5\delta$. Summing over t and using $\sum_{t=2}^T (t-1)^{-1/2} \leq 2\sqrt{T}$, we obtain

$$\begin{aligned}
 &\sum_{t \in [T] \setminus \mathcal{E}_T} \{ \mathbb{1}(a_t = \hat{a}_t) (\mathbb{E}_{t-1}[y_{*,t}] - \mathbb{E}_{t-1}[y_{a_t,t}]) \} \\
 &\leq \frac{32\sigma\kappa_{\max,x}\kappa_{\max,y}}{\tilde{\sigma}_{\min,z}p} \sqrt{2(d_x + d_{h_x})(d_y + d_{h_y})T \log\left(\frac{2MNT^2}{\delta}\right)}.
 \end{aligned}$$

Finally, by Lemma D.3 and a Hoeffding–Azuma inequality, with probability at least $1 - \delta$,

$$\begin{aligned}
 &\sum_{t \in [T] \setminus \mathcal{E}_T} \mathbb{1}(a_t \neq \hat{a}_t) (\mathbb{E}_{t-1}[y_{*,t}] - \mathbb{E}_{t-1}[y_{a_t,t}]) \\
 &\leq 2 \sum_{t=1}^T \mathbb{1}(a_t \neq \hat{a}_t) \\
 &\leq 2 \sum_{t=1}^T \mathbb{P}_{t-1}(a_t \neq \hat{a}_t) + \sqrt{2T \log\left(\frac{2}{\delta}\right)} \\
 &\leq \sqrt{2T \log\left(\frac{2}{\delta}\right)} + 4\sqrt{T} + 2\delta.
 \end{aligned}$$

Putting all components together, we conclude that with probability at least $1 - 6\delta$,

$$\begin{aligned}
 \mathbf{Regret}(T) &\leq \frac{1024MN(MN - 2 + p)\tilde{\sigma}_{\max,z}^2}{3(1-p)\tilde{\sigma}_{\min,z}^2} \log\left(\frac{2MNT^2}{\delta}\right) \\
 &\quad + \frac{32\sigma\kappa_{\max,x}\kappa_{\max,y}}{\tilde{\sigma}_{\min,z}p} \sqrt{2(d_x + d_{h_x})(d_y + d_{h_y})T \log\left(\frac{2MNT^2}{\delta}\right)} \\
 &\quad + \sqrt{2T \log\left(\frac{2}{\delta}\right)} + 4\sqrt{T} + 2\delta.
 \end{aligned}$$

which concludes the proof. $\square$

D. Technical Lemmas

Lemma D.1 (Sub-Gaussian martingale inequality). *Let $(X_t)_{t \geq 0}$ be a martingale adapted to a filtration $(\mathcal{F}_t)_{t \geq 0}$. Suppose that, for every $\lambda \in \mathbb{R}$ and all $t \geq 1$,*

$$\mathbb{E}[\exp(\lambda(X_t - X_{t-1})) \mid \mathcal{F}_{t-1}] \leq \exp\left(\frac{\lambda^2 \sigma_t^2}{2}\right)$$

for some $\sigma_t > 0$. Then, for all $a \geq 0$ and $T \geq 1$,

$$\mathbb{P}(|X_T - X_0| \geq a) \leq 2 \exp\left(-\frac{a^2}{2 \sum_{t=1}^T \sigma_t^2}\right).$$

Consequently, with probability at least $1 - \delta$ for any $\delta \in (0, 1)$,

$$|X_T - X_0| \leq \sqrt{2 \sum_{t=1}^T \sigma_t^2 \log \frac{2}{\delta}}.$$

Theorem D.2 (Freedman inequality). *Let $\{\mathcal{F}_k\}_{k \geq 0}$ be a filtration. Let $\{Y_k\}_{k \geq 0}$ be a real-valued martingale adapted to $\{\mathcal{F}_k\}$ with $Y_0 = 0$, and define the martingale differences $X_k := Y_k - Y_{k-1}$. Assume that there exists $R > 0$ such that*

$$X_k \leq R \quad \text{almost surely for all } k \geq 1.$$

Define the predictable quadratic variation process

$$W_k := \sum_{j=1}^k \mathbb{E}[X_j^2 \mid \mathcal{F}_{j-1}], \quad k \geq 1.$$

Then for all $x \geq 0$ and all $\sigma^2 > 0$,

$$\mathbb{P}(\exists k \geq 0 : Y_k \geq x \text{ and } W_k \leq \sigma^2) \leq \exp\left(-\frac{x^2/2}{\sigma^2 + Rx/3}\right).$$

Lemma D.3 (Bound for the probability of exploration). *For each t , let $\hat{a}_t = \operatorname{argmax}_{(i,j) \in [M] \times [N]} \tilde{\mathbf{x}}_i^\top \hat{\Phi}_{t-1} \tilde{\mathbf{y}}_j$ denote the maximizing action based on the estimator $\hat{\Phi}_t$. Then the action a_t chosen by the resampling scheme in Algorithm 1 satisfies*

$$\sum_{t=1}^T \mathbb{P}(a_t \neq \hat{a}_t) \leq 2\sqrt{T} + \delta.$$

Proof. Let $a_t = (i_t, j_t)$, $\hat{a}_t = (\hat{i}_t, \hat{j}_t)$, and $\tilde{a}_t = (\tilde{i}_t, \tilde{j}_t)$. For any fixed $t \in [T]$, we decompose

$$\begin{aligned} \mathbb{P}(a_t \neq \hat{a}_t) &= \mathbb{P}(\{\tilde{a}_t = a_t\} \cap \{a_t \neq \hat{a}_t\}) + \mathbb{P}(\{\tilde{a}_t \neq a_t\} \cap \{a_t \neq \hat{a}_t\}) \\ &\leq \mathbb{P}(\{\tilde{a}_t = a_t\} \cap \{a_t \neq \hat{a}_t\}) + \mathbb{P}(\tilde{a}_t \neq a_t). \end{aligned}$$

For the first term, we have

$$\begin{aligned} \mathbb{P}(\{\tilde{a}_t = a_t\} \cap \{a_t \neq \hat{a}_t\}) &= \mathbb{P}(\{\tilde{a}_t = a_t\} \cap \{\tilde{a}_t \neq \hat{a}_t\}) \\ &\leq \mathbb{P}(\tilde{a}_t \neq \hat{a}_t) \\ &= \mathbb{E}[\mathbb{P}(\tilde{a}_t \neq \hat{a}_t \mid \mathcal{F}_{t-1})] \\ &\leq \frac{1}{\sqrt{t}}. \end{aligned}$$

where the last inequality follows from the exploration distribution of Algorithm 1, which assigns total probability mass $1/\sqrt{t}$ to actions other than $\hat{a}_t$. For the second term, the resampling success guarantee gives

$$\mathbb{P}(\tilde{a}_t \neq a_t) \leq \frac{\delta}{(t+1)^2}.$$

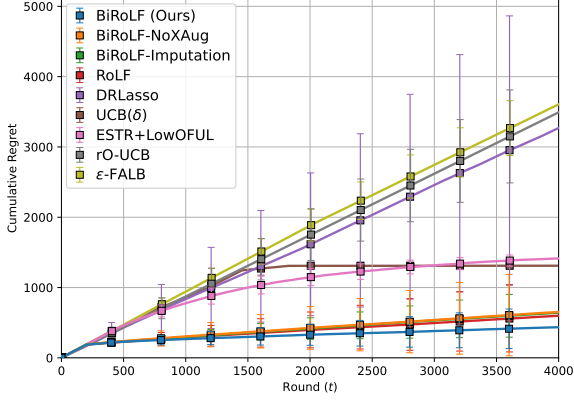
Therefore, combining the two terms yields

$$\mathbb{P}(a_t \neq \hat{a}_t) \leq \frac{1}{\sqrt{t}} + \frac{\delta}{(t+1)^2}.$$

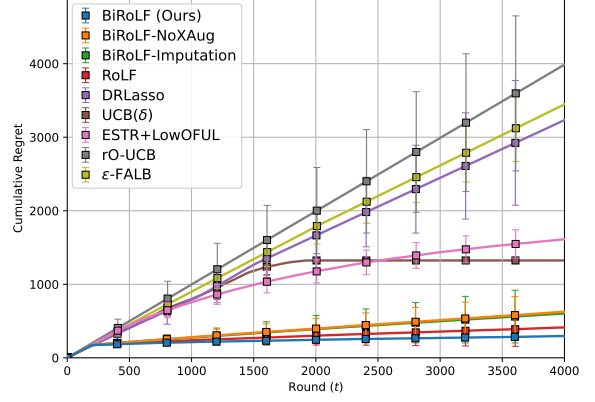
Finally, summing over all $t \in [T]$, and using $\sum_{t=1}^T t^{-1/2} \leq 2\sqrt{T}$ and $\sum_{t=1}^{\infty} \delta/(t+1)^2 \leq \delta$ yields the result. $\square$

E. Additional Experiments

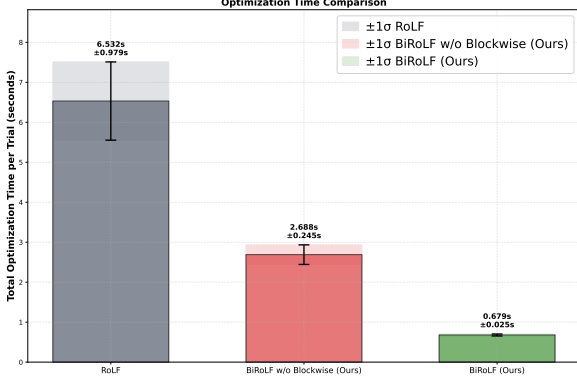
This appendix provides the full experimental results omitted from the main text. The main text focuses on the optimization-time comparison, since the main empirical advantage of BiRoLF is computational. Here we additionally report cumulative regret, total optimization time, and wall-clock time for all four synthetic cases and for a semi-synthetic MovieLens experiment. Unless otherwise stated, the synthetic experiments use the same parameters as in Section 7: $M = N = 20$, $d_x^* = d_y^* = 15$, $d_x = d_y = 7$, horizon $T = 4000$, and five independent trials. The four cases follow the row-space relationships in Figure 2. We report the mean cumulative regret with one-standard-deviation error bars. For timing, optimization time denotes the total time spent solving the main ℓ_1 -regularized estimator over one trial. Wall-clock time denotes the total running time of one trial. The timing plots compare RoLF, BiRoLF w/o Blockwise, and BiRoLF, since these three methods isolate the effect of Kronecker linearization and the exact blockwise solver. For the semi-synthetic experiment, we use MovieLens rating data to construct a recommendation-style bilinear bandit instance. Movie and user descriptors are used as observed side information, and additional latent dimensions are appended according to the same four case definitions. We fit a bilinear reward model to the observed ratings and use the fitted reward matrix as the ground-truth mean reward matrix for bandit simulation. This keeps the bandit feedback evaluation controlled while incorporating structure from a real recommendation dataset. The results support the same conclusion across synthetic and semi-synthetic settings. BiRoLF achieves cumulative regret comparable to RoLF, which is the strongest partially observable baseline based on Kronecker linearization. At the same time, BiRoLF substantially reduces optimization time, and also improves wall-clock time relative to RoLF. The ablations show that removing the user-side augmentation or removing the DR correction degrades regret, which confirms that both components are useful under partial observability. The fully observed bilinear baselines are not designed for latent reward components and generally perform worse in these partially observable settings.



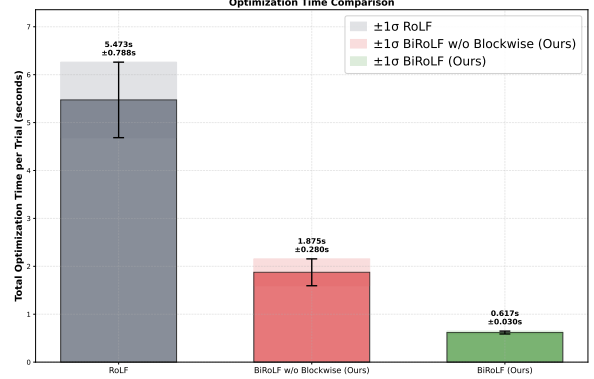
(a) Case 1: Cumulative regret.



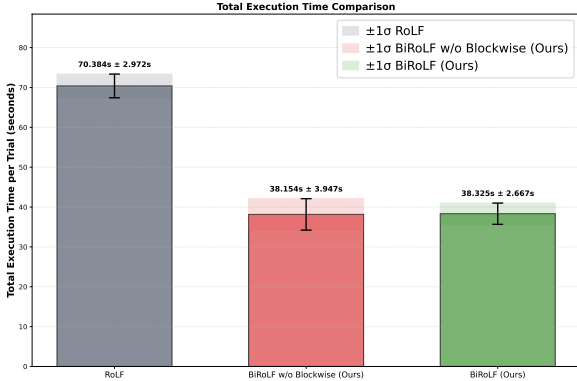
(b) Case 2: Cumulative regret.



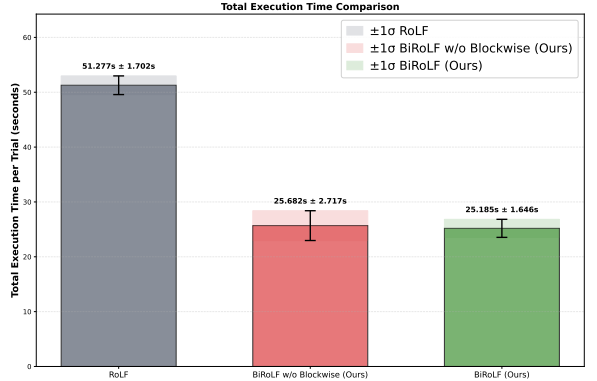
(c) Case 1: Optimization time.



(d) Case 2: Optimization time.

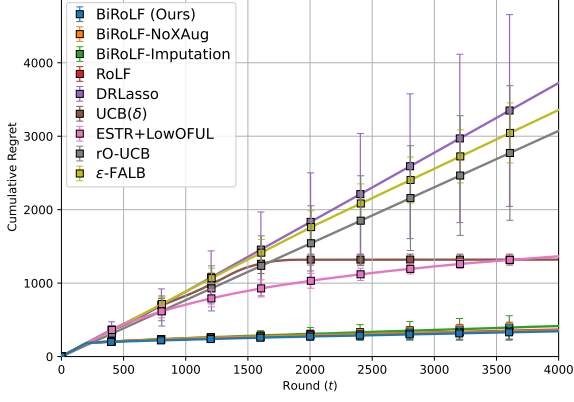


(e) Case 1: Wall-clock time.

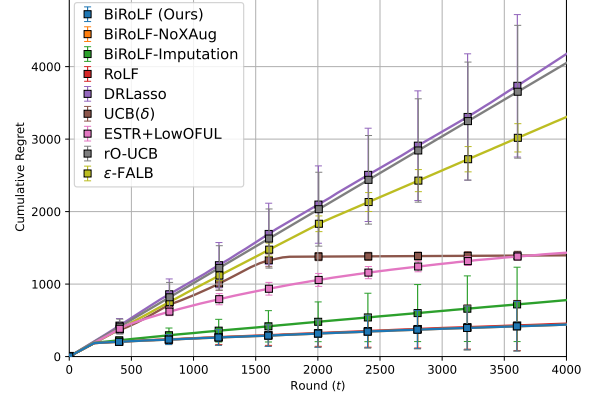


(f) Case 2: Wall-clock time.

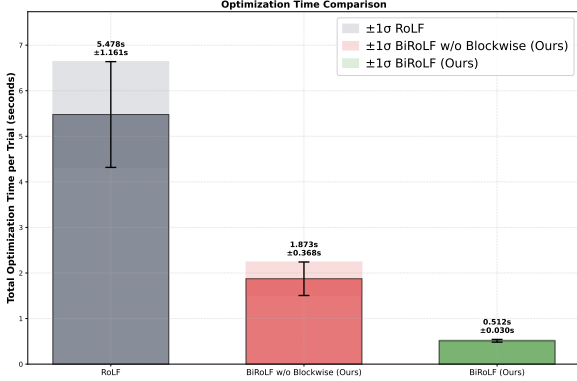
Figure 5. Synthetic experiment (Cases 1-2).



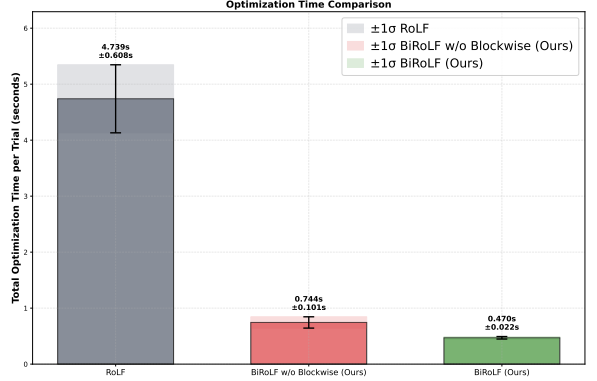
(a) Case 3: Cumulative regret.



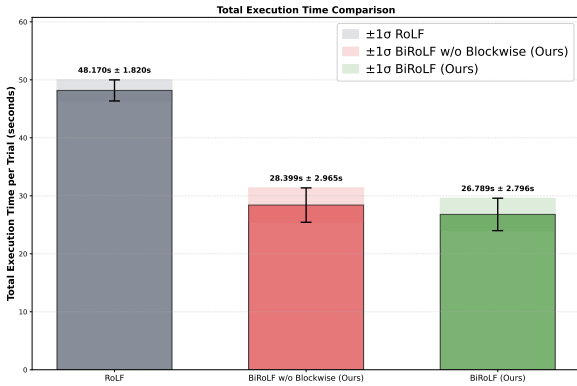
(b) Case 4: Cumulative regret.



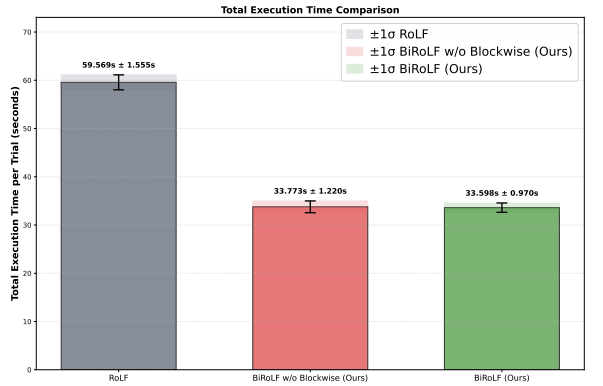
(c) Case 3: Optimization time.



(d) Case 4: Optimization time.

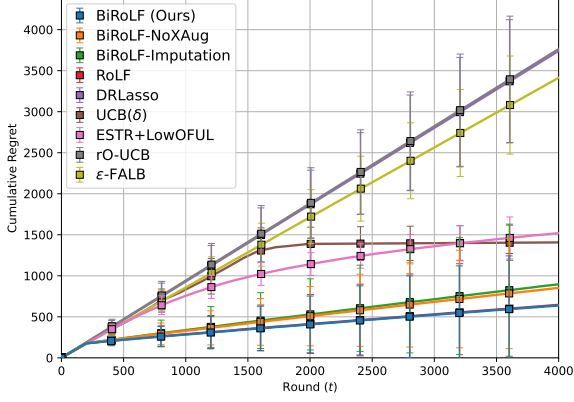


(e) Case 3: Wall-clock time.

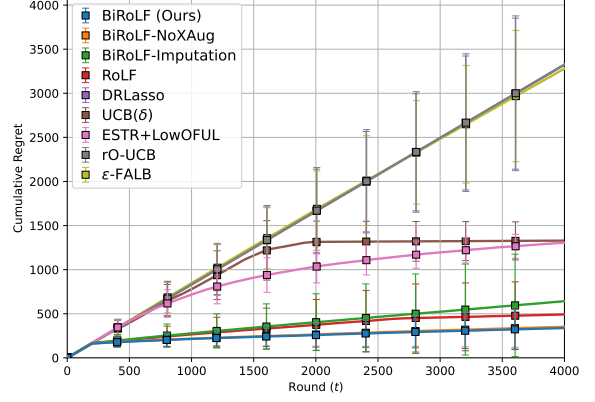


(f) Case 4: Wall-clock time.

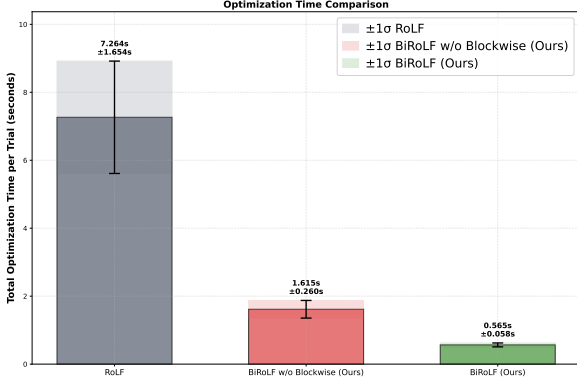
Figure 6. Synthetic experiment (Cases 3-4).



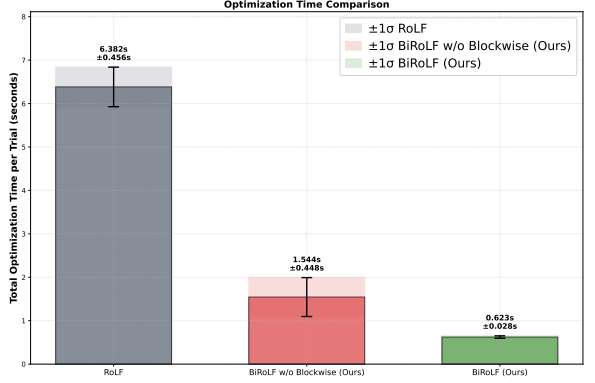
(a) Case 1: Cumulative regret.



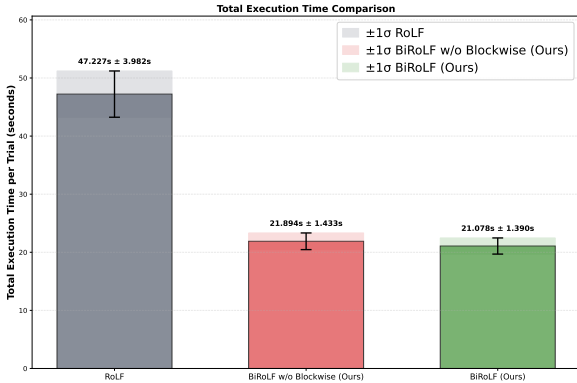
(b) Case 2: Cumulative regret.



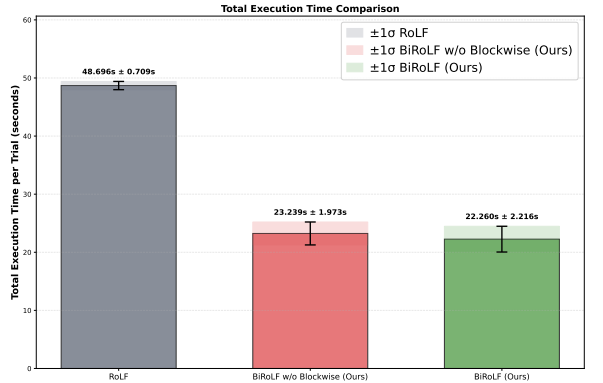
(c) Case 1: Optimization time.



(d) Case 2: Optimization time.

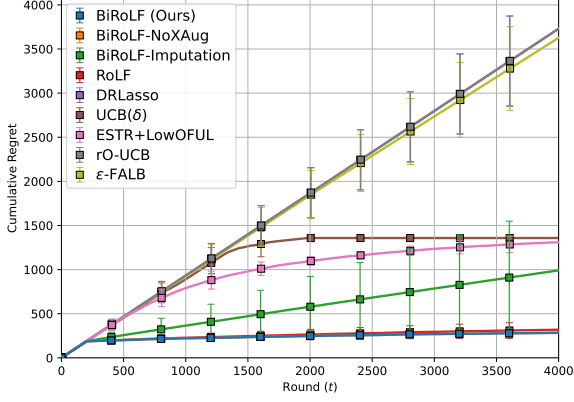


(e) Case 1: Wall-clock time.

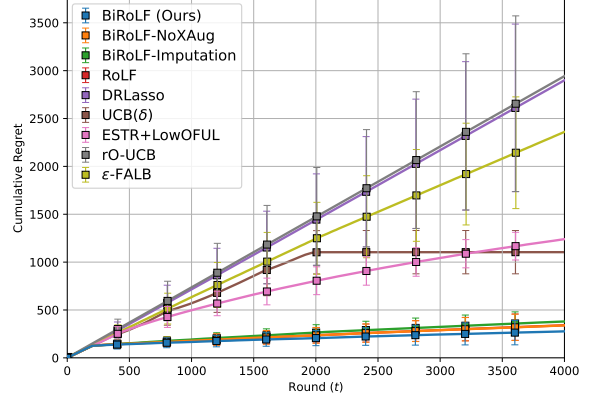


(f) Case 2: Wall-clock time.

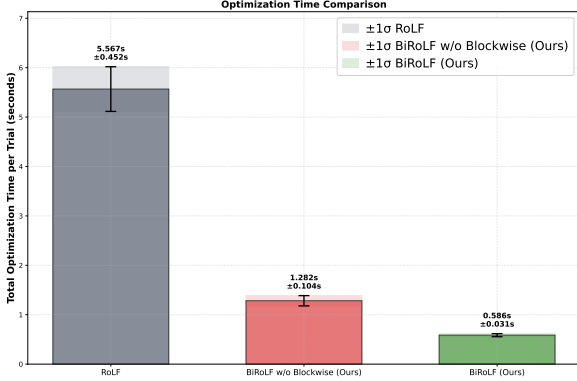
Figure 7. Semi-synthetic experiment (Cases 1-2).



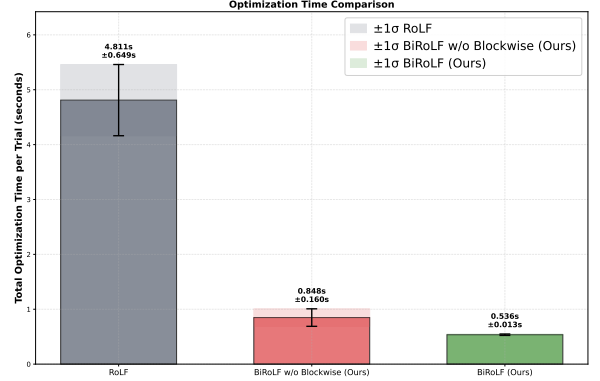
(a) Case 3: Cumulative regret.



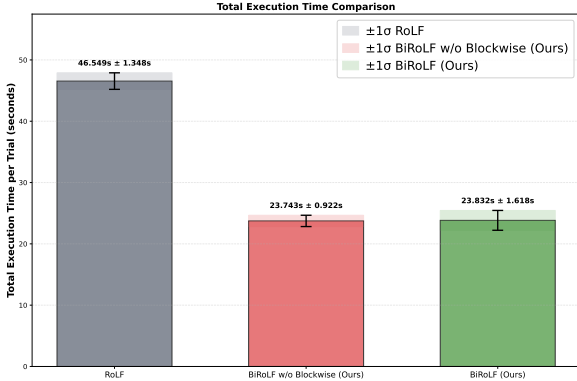
(b) Case 4: Cumulative regret.



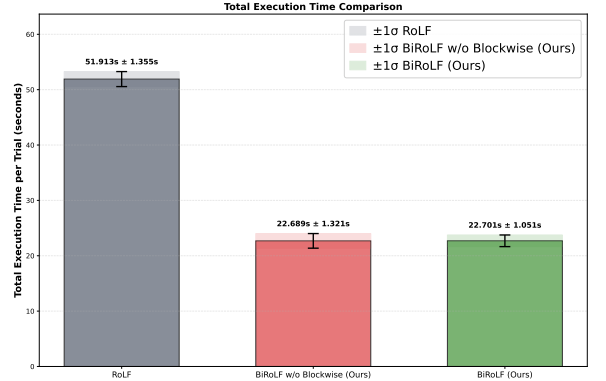
(c) Case 3: Optimization time.



(d) Case 4: Optimization time.



(e) Case 3: Wall-clock time.



(f) Case 4: Wall-clock time.

Figure 8. Semi-synthetic experiment (Cases 3-4).

F. Limitations and Future Directions

This paper focuses on persistent partial observability. At the start of learning, the learner is given a fixed two-sided action pool and fixed observed user and item descriptors, while some reward-relevant components are never observed. Thus, our goal is not online feature acquisition, where missing features may later be queried or revealed. Instead, we study robust learning from bandit feedback under a fixed partially observed representation. This is a natural extension of the standard fixed-pool bilinear bandit benchmark, but it also determines the scope of our guarantees.

The main limitation is that our analysis relies on fixed observed feature matrices. The orthogonal-complement bases and the blockwise DR-Lasso decomposition are both constructed from the row spaces of these matrices and are kept fixed throughout learning. If user or item contexts change over time, the current augmentation would no longer give a fixed representation, and the estimator and regret analysis would have to be modified. The framework also assumes homogeneous observability on each side. Users share one observed user-side feature space, and items share one observed item-side feature space. Thus, it does not cover settings where different users or items reveal different subsets of descriptors. Extending the method to such settings would require user- or item-specific observation models and complement bases.

We view dynamic contexts and heterogeneous observability as the most direct next steps. Another useful direction is to combine the present augmentation with low-rank estimation. Although a low-rank Θ_* is allowed by our model, BiRoLF does not use a low-rank factorization. Instead, it exploits sparsity in the orthogonal-complement blocks of the effective parameter. Using both structures would require a different estimator, since the orthogonal augmentation, DR imputation, and low-rank constraint would need to be handled jointly.